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HY PULMONARY



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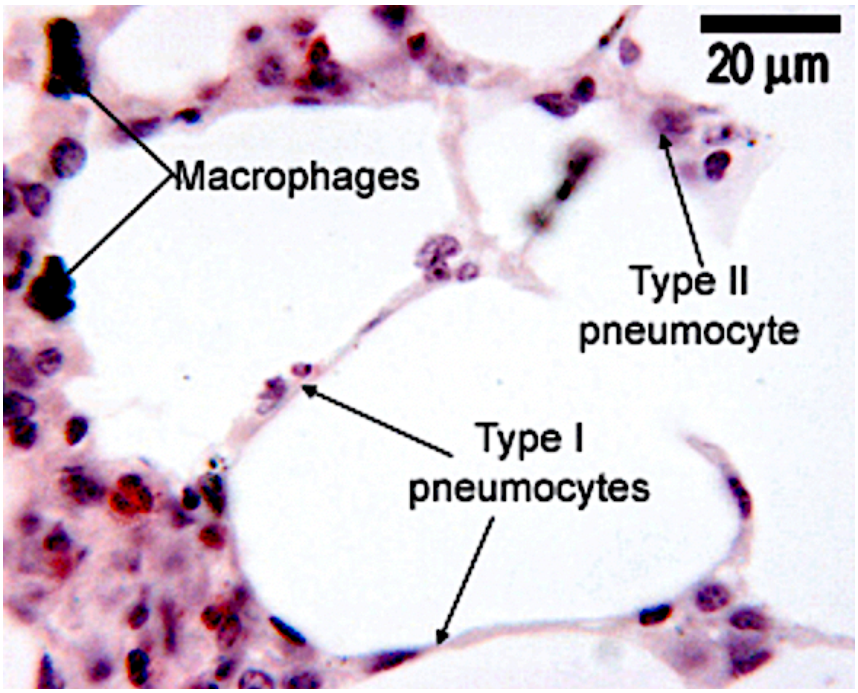
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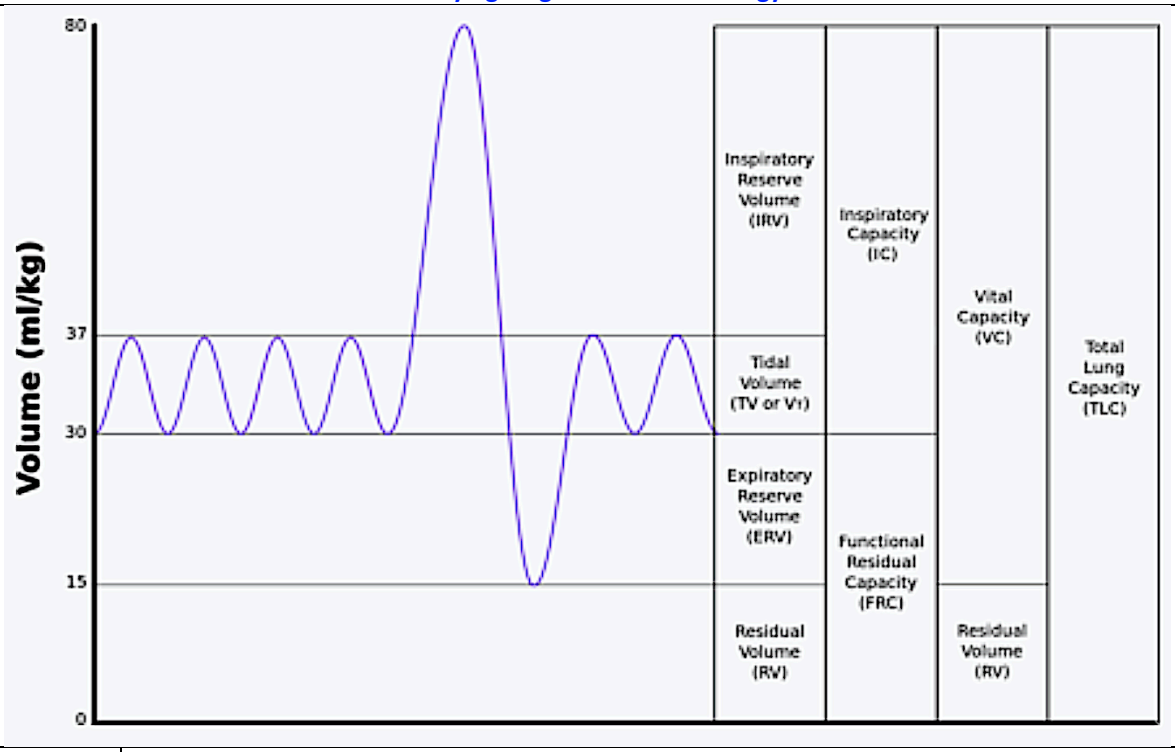
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HY Pulmonary – by Dr Mike Mehlman

HY Pulmonary cell types + path points	
Type I pneumocyte	- Simple squamous; flat and long in appearance; carries out gas exchange; 95% surface area of the alveoli. Although Step 1 is P/F now, USMLE is known to occasionally give a fuck and show alveolar histo, so just be able to identify the basic cell types.  - You say, "No idea what I'm looking at." Notice how the type I pneumocytes appear long and flat. The type II pneumocytes and macrophages are bulkier. - Emphysema is destruction of alveolar surface area (i.e., mostly destruction of type I pneumocytes) → ↓ surface area for gas exchange → chronic CO_2 retention → chronic respiratory acidosis (high bicarb to compensate).
Type II pneumocyte	- Two functions on USMLE: - 1) Stem cells of the lung that proliferate in response to damage (i.e., if Q says type I pneumocytes are damaged + ask what cell is responsible for replenishing them, the answer is type II pneumocyte). - 2) Contain specialized organelles called lamellar bodies that produce surfactant. - In neonatal respiratory distress syndrome (NRDS), there are ↓ lamellar bodies.
Pseudostratified columnar epithelial cells with cilia	- Present throughout the respiratory tree from the nasopharynx down to the respiratory bronchioles. - USMLE wants you to know this is the primary cell that's fucked up in smoking. - Even though function of other pulmonary cell types will also be diminished, this is specifically the answer on the NBME. - In other words, you might think that activity of this cell type might be ↑ in smoking in order to ↑ particulate clearance, but activity is actually impaired.
Alveolar macrophage	- Phagocytose pathogens and help clear particulates out of the lungs. - The answer for the cell that TB lives within. The macrophages then interact with CD4+ T cells. The latter stimulate macrophage killing ability + also initiate granuloma formation. - The answer for the cell that produces 1α-hydroxylase in sarcoidosis, resulting in vitamin D activation, ↑ 1,25-D3, and hyperCa^{2+} due to ↑ small bowel absorption.

	- The answer on USMLE for the cell that initiates pulmonary fibrosis. For example, they'll show image of ferruginous body in asbestosis + ask which cell initiates pulmonary fibrosis in this patient → answer = macrophage. - Activity is ↓, not ↑, in smokers. Similar to the pseudostratified columnar ciliated cells, even though we might expect activity to ↑ in order to clear out particulates, the latter actually impair the function of alveolar macrophages.
Basement membranes	- NBME Path questions are obsessed with you knowing integrity of basement membranes must be maintained for restoration of normal pulmonary architecture following infection or trauma. - Q might say patient has pulmonary abscess treated successfully + year later there is lesion still seen; why? → answer = "failure of maintenance of basement membranes." - Or, the Q says patient has pneumonia + simply asks what enables for there to be complete restoration of pulmonary architecture → answer = "maintenance of basement membranes." It's important I'm repetitive on this point because it shows up repeatedly on the Step 1 NBME forms.
Kulchitsky cell	- Aka neuroendocrine cell of the respiratory tract. - This is the cell that is the origin for small cell lung cancer.

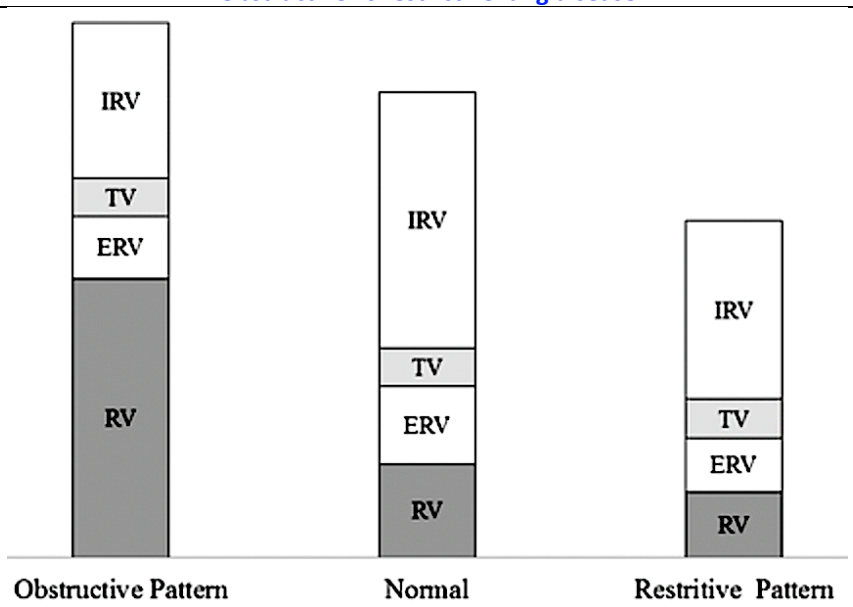
Annoying lung volume terminology



FEV1	- Forced Expiratory Volume in 1 second. - The amount of air one can forcefully exhale in one second following a maximal inspiration.
FVC (VC)	- Forced Vital Capacity. - The total volume of air one can maximally exhale following a maximal inspiration.
FEV1/FVC	- Ratio used to differentiate between obstructive and restrictive lung disease.
TLC	- Total Lung Capacity. - The total amount of air in the lungs following maximal inspiration.
TV	- Tidal Volume. - The amount of air inhaled + exhaled during normal/casual breathing.
IC	- Inspiratory Capacity. - The maximum amount of air that can be inhaled after a normal/casual exhalation.

IRV	- Inspiratory Reserve Volume. - Additional amount of air that can be inhaled after a normal inhalation.
ERV	- Expiratory Reserve Volume. - Additional amount of air that can be exhaled after a normal exhalation.
FRC	- Functional Residual Capacity. - Volume of air remaining in the lungs after a normal exhalation.
RV	- Residual Volume. - Volume of air remaining in the lungs after a maximal exhalation.
DLCO	- Not a lung volume term. But you'll handle it. - Diffusion capacity of the lungs for carbon monoxide. Reflects extent to which gas exchange is occurring across the pulmonary capillaries. - For USMLE purposes, it is decreased in every pathologic lung condition, but increased in asthma. This is thought to be due to a combination of transient ↑ in blood flow through the pulmonary capillaries + additional hyperinflation during acute episodes. - In other words, if you get a big paragraph Q where you're not sure what's going on, but they say DLCO is reduced, you can say, "Cool, not asthma." Or if they say DLCO is high, you can just select asthma without even reading the question basically.

Obstructive vs restrictive lung disease

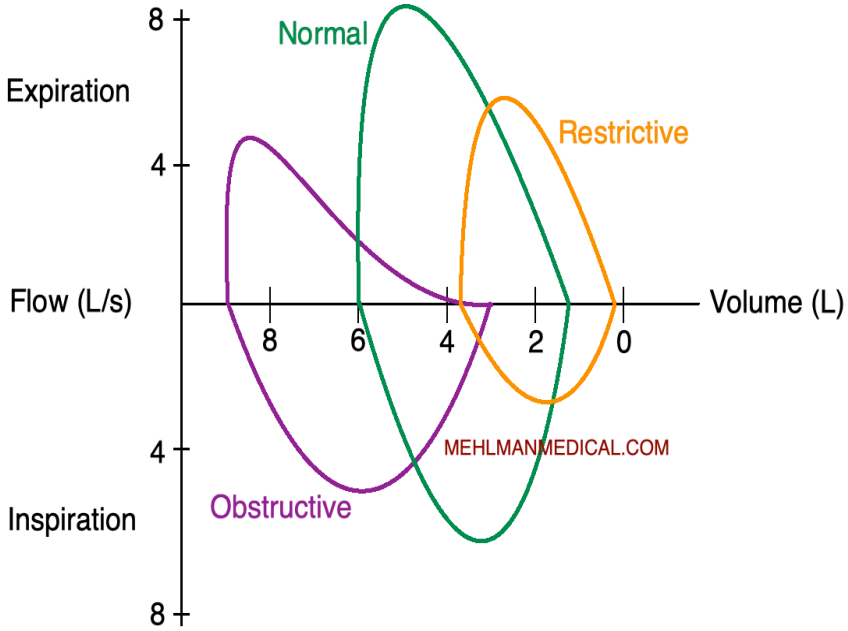


	Obstructive	**Restrictive**
HY conditions	- Asthma. - COPD (chronic bronchitis + emphysema). - **Old age (i.e., healthy person age 70 vs 20).**	- Pulmonary fibrosis due to any cause (i.e., idiopathic; radiation; drugs; auto immune disease). - Pneumoconioses.
FEV1	↓	↓/↔
FVC	↓	↓
FEV1/FVC	**↓ (<70%; normal is 70%)**	**↑/↔ (>70-80%)**
TLC	↑	↓
FVC (VC)	↔/↓	↓
TV	↔	↔/↓
IC	↔/↓	↓
IRV	↔/↓	↓
ERV	↓	↓
FRC	↑ (due to air trapping)	↓ (due to ↓ lung compliance)
RV	↑ (due to air trapping)	↓ (due to ↓ lung compliance)

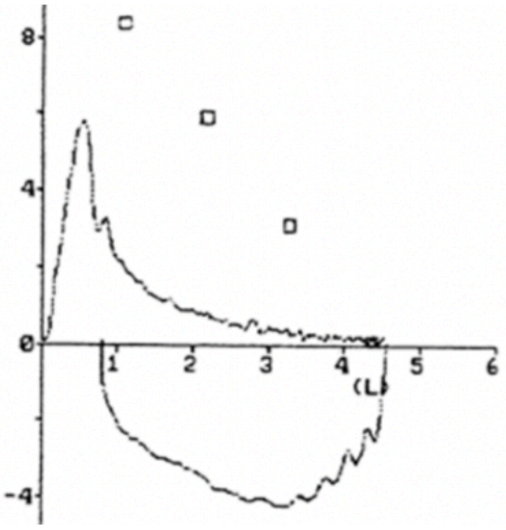
DLCO	↓ (but ↑ in asthma on USMLE)	↓ (↔ if neuromuscular)

- The reason FEV1/FVC is ↑/↔ in restrictive, as compared to ↓ in obstructive, is because of **radial traction,** which is a "stickiness" on the outside of the airways (such as from fibrosis) that prevents them from closing as rapidly as in obstructive.
- If you feel iffy on lung volumes, a general point to remember is that they're basically all decreased in restrictive due to ↓ compliance, but various ones are ↑ in obstructive because air can't get out.
- Lifetime inhalation of particulates (i.e., non-smoker simply living in a city) causes an accreted effect on the lungs akin to very slow-onset COPD. So if the USMLE asks for lung volumes in a healthy 70-year-old versus a 20-year-old, select arrows for the former consistent with obstructive lung disease.

Flow-volume loop garbage



- Don't freak out about the above flow-loops. For USMLE, I'd say the most important point is that you know the expiratory component of the obstructive curve (purple) has a scooped-out/concave shape. This is because obstruction causes the flow rate to be significantly reduced throughout the expiration.



- What USMLE will do is give you a real bootleg Windows 95-type spirometry curve as per above in a 34-year-old + no family Hx + 5-year smoking Hx. You look at curve and say, "No idea what I'm looking at, but

top of curve is scooped-out/concave, and I know that means obstructive. So if I'm guessing between COPD or asthma here, which are both obstructive, I'll choose asthma as more likely over COPD in young patient."		
	Obstructive	**Restrictive**
Shape	- **Scooped-out/concave** appearance.	- Smaller and more symmetric.
Expiratory flow rate	- Airway obstruction causes big reduction in flow rate, which is why it drops precipitously and results in scooped curve.	- Peak expiratory flow rate relatively preserved, but overall size of the loop is reduced due to less volume in the lungs on maximum inspiration.
Inspiratory flow rate	- Less affected compared to expiratory curve. Can be normal or reduced.	- **Significantly reduced** due to diminished pulmonary compliance.
Volume	- Wider curve, due to increased TLC from air trapping.	- Narrower curve, due to diminished TLC from reduced compliance.

Shunt versus dead space

- When we discuss V/Q mismatch, V = ventilation; Q = perfusion.
- ↓ V/Q (ventilation/perfusion) = reduced ventilation relative to perfusion.
- ↑ V/Q = reduced perfusion relative to ventilation.
- **Shunt** = ↓ V/Q → refers to almost all lung conditions on USMLE.
- **Dead space** = ↑ V/Q → refers to basically just pulmonary embolism on USMLE.

- If you hear the term "shunt" used in isolation, this refers interchangeably to "pathologic shunt," where ↓ V/Q occurs to the point that the patient's arterial oxygen becomes decompensated. In other words, the reason why a patient has ↓ O2 sats is because there's some sort of lung problem where aeration of the alveoli is ↓. But there's nothing wrong with perfusion (i.e., blood flow isn't cut off to the lungs).
- ↓ V/Q can also refer to the physiologic (i.e., normal) shunt we see at the lung bases, where V/Q is 0.6. This is because despite both ventilation and perfusion ↑ apex → base, gravity pulls blood down more greatly than air, so perfusion ↑↑ toward the bases, whereas ventilation only ↑. So V/Q is ↓ at the bases (shunt).
- Areas of lung with ↓ V/Q attempt to compensate via hypoxic vasoconstriction, where blood vessels constrict in order to redirect (i.e., shunt) blood to better-ventilated areas. In other words, if an area has low ventilation, constriction of blood vessels in that area lowers Q in order to increase the V/Q ratio and prevent wasting of blood flow.

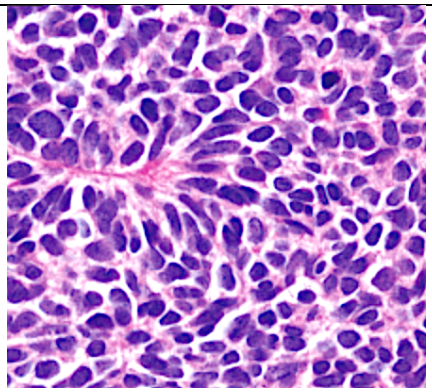
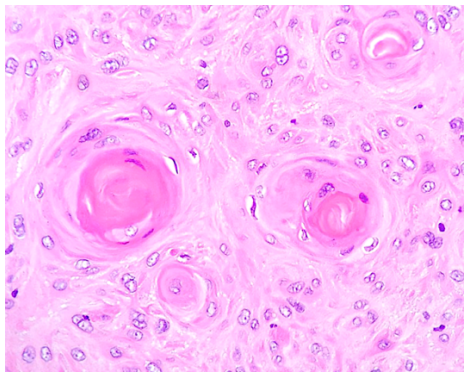
- If you hear the term "dead space" used in isolation, this refers interchangeably to "pathologic dead space," where ↑ V/Q occurs to the point that the patient's arterial oxygen becomes decompensated. In other words, the reason why a patient has ↓ O2 sats is because blood flow is cut off to the lungs. This will be PE (and air, fat, amniotic fluid emboli) on USMLE because the clot that embolizes to the pulmonary arteries compromises blood flow to the pulmonary tissue.
- ↑ V/Q can also refer to dead space processes that are non-pathologic:
    - *Alveolar* dead space = natural, physiologic ↑ V/Q within the alveoli, where some areas of lung receive more ventilation than perfusion – i.e., the apices (V/Q of ~3.0) compared to the bases (~0.6). This is because gravity pulls blood down more than air, so at the apices of the lungs, there's relatively less blood flow in comparison to how much those areas are ventilated.
    - *Anatomic* dead space refers to parts of the respiratory tree that are naturally ventilated but do not partake in gas exchange, such as the trachea, bronchi, and terminal bronchioles. These areas do not receive perfusion where gas exchange occurs for respiratory purposes, despite being ventilated by air moving in and out.
    - *Physiologic* dead space is the sum of anatomic and alveolar dead space.

- To understand shunt vs dead space better, both involve a right-to-left mixing of deoxygenated blood (right) with oxygenated blood (left). Shunt is a R-to-L mixing where "R" is created by areas of alveoli not receiving enough oxygen; dead space is a R-to-L mixing where "R" is due to ↓ perfusion to areas of lung.
- An area of lung under-ventilated or -perfused could be said to represent a "zero" in terms of the oxygenation it contributes to the blood. That zero is mixed in with all of the other areas of normal lung. This means *the average* of all areas of lung cannot achieve normal oxygenation because that zero is mixed in. This is why arterial O2 will be decreased when we have pathologic shunt or dead space.

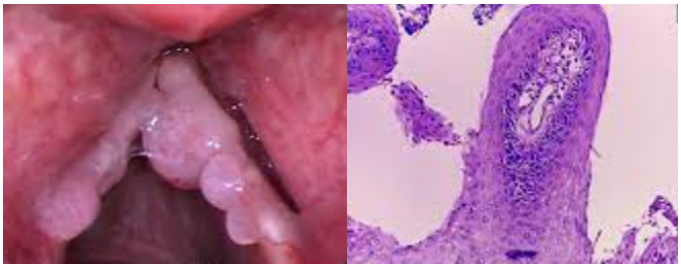
- In both pathologic shunts (e.g., COPD, pneumonia, asthma) and dead space (PE), there are still areas of lung that have normal V/Q. Administering oxygen to these patients will have differing effect, where shunt is less effectively remedied often due to the extent of hypoxic vasoconstriction, which causes a high fraction of blood to bypass normal V/Q areas altogether, even though a small fraction is diverted appropriately to them. The net result is oxygen administration still fails to increase PaO2 in shunt. In dead space, where hypoxic vasoconstriction doesn't occur because ventilation is normal, administered oxygen to normal V/Q areas helps because it comes into contact with more blood overall.
- This can essentially be simplified as: dead space by definition is a *primary* ↓ in perfusion; shunt causes a *secondary* ↓ in perfusion. But we don't call the secondary ↓ in perfusion "dead space." Hypoxic vasoconstriction due to shunt is a mere compensatory response to ↓ ventilation, but it happens to be the case that the blood that is shunted away from the lungs can be greater in volume than the primary ↓ in perfusion in the setting of a pathologic dead space.

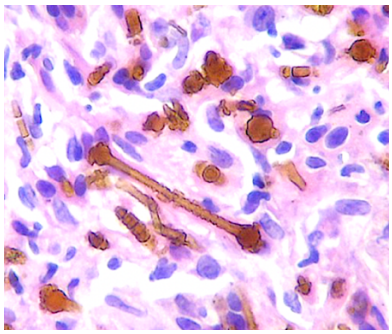
Alveolar-arterial (A-a) gradient stuff
- A is alveolar oxygen; a is arterial oxygen. - In settings where the patient's arterial oxygen is low, the A-a gradient (normally 5-10 mmHg) tells us whether there's an actual lung pathology or if the patient is merely hypoventilating. In other words, - ↑ **A-a gradient = the lungs are fucked up** = the patient is breathing normally but there is a problem such as impediment of gas exchange, poor lung compliance, or pathologic shunt or dead space. The patient has low arterial oxygen despite sufficient respirations, where alveolar pO2 is normal in unaffected areas of lung. The normal "A" in unaffected areas of lung minus the low "a" in the blood makes A-a high. - **Normal A-a gradient = the lungs are normal** = patient is just not breathing enough – i.e., quality and/or quantity of respirations is insufficient. This refers to **opioids, heroin, benzos, and barbiturates** because these agents cause respiratory depression. Normal A-a gradient is also sometimes seen in patients on ventilators. The patient has low arterial oxygen merely because of insufficient respirations, where alveolar oxygen is low throughout the lungs. The low "A" throughout the lungs is due to poor respirations, minus the low "a" in the blood, making A-a not elevated. - The NBME doesn't expect you to know the exact numbers for A-a gradients. The Q might give you answer choices where A-a gradient is either 40 or 10, where you have to say, "Well clearly the answers with 40 are high and the ones with 10 are normal." So if the Q is a patient who has opioid toxicity with low O2 sats, you'd select the answer choice with 10 for A-a rather than 40. Not dramatic. - USMLE also wants you to know that the mechanism for a patient's hypoxemia in pulmonary edema is "high A-a gradient"; this makes sense, as the transudate in the alveolar spaces impedes gas exchange, but the patient's respirations are otherwise fine. - Old age in healthy persons causes: ↑ A-a gradient, ↓ arterial pO2, ↑ TLC. This combo is answer on NBME – i.e., even if people are non-smokers, as we live out life inhaling particulates from car exhaust, etc., the lungs experience obstructive changes with age (essentially "very slow COPD"). - Choose normal A-a gradient in patients on ventilators who have ↑ pCO2. The latter tells us the arterial pO2 is low because of insufficient respirations. - My biggest advice is to not automatically assume "high A-a gradient" for every scenario of low O2 sats. The mistake I see students make is they just think "any issue causing low O2 sats = A-a gradient must be high."

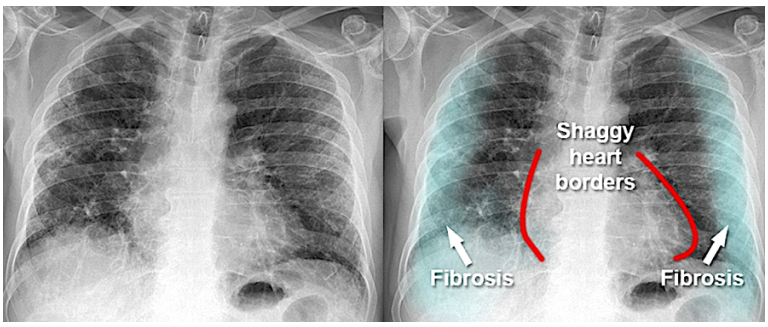
HY pulmonary / respiratory tract cancers for USMLE	
Small cell bronchogenic carcinoma	- Occurs centrally in the lung (i.e., hilar / medially). Smoking biggest risk factor. - Paraneoplastic syndromes exceedingly HY: 1) SIADH (ADH secretion); 2) Cushing syndrome due to ACTH secretion; 3) Lambert-Eaton syndrome (production of antibodies against presynaptic voltage-gated calcium channels); 4) Cerebellar dysfunction / ataxia (anti-Hu/-Yo antibodies). - Small basophilic (purple) cells. NBME can describe these cancer cells as "twice the size of lymphocytes."

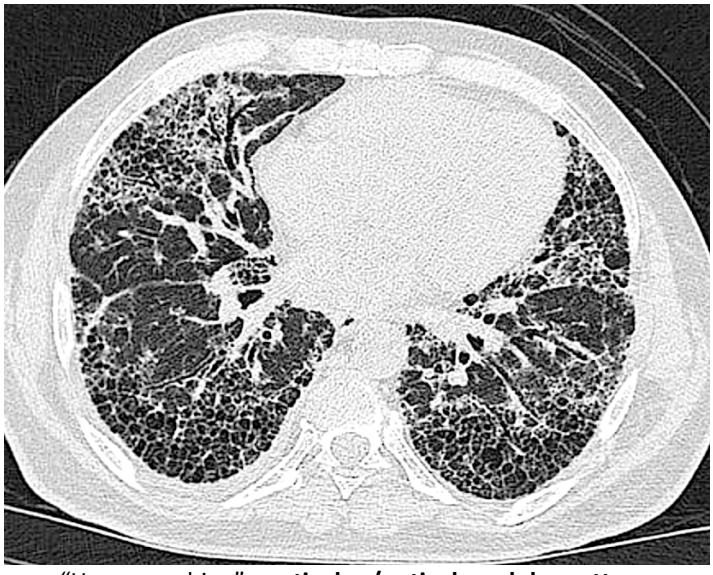
	 - Treatment on 2CK is "chemotherapy." Do not do surgery. I believe this is the only time I've seen straight-up "chemotherapy" as correct answer on NBME.
Squamous cell carcinoma	- Smoking biggest risk factor. Similar to small cell, occurs centrally in the lung (i.e., hilar / medially). In other words, the two cancers that start with an "**ssss**" sound for "**c**entral" (i.e., **S**mall cell and **S**quamous cell) are **C**entral. - Stains positive for keratin; has keratin pearls on histo (pink circles).  - Can cavitate → if USMLE gives you lung cancer with a cavitation (i.e., cavity/hole), the answer is squamous cell.  - Highest yield point is that it secretes PTHrp (parathyroid hormone-related peptide). This acts like PTH and increases calcium / decreases phosphate, but this is not the same as PTH. Endogenous PTH is *suppressed* due to negative feedback from high calcium. In other words, choose a down arrow for PTH in squamous cell carcinoma of the lung.
Adenocarcinoma of the lung	- Adenocarcinoma means cancer of glands. If the Q says something about biopsy showing glandular morphology, you know they're talking about adenocarcinoma.

	- Likewise, the Q can say the cancer does *not* show any glandular morphology, and you know that adenocarcinoma is wrong. - The answer on USMLE for lung cancer in a non-smoker; classically "female non-smokers," but I've seen NBME Qs with this in men. - Normal ground/Earth radiation due to radon is accepted cause of lung cancer in non-smokers. There is NBME Q where they mention non-smoker living in a basement and develops lung cancer. The correlation is probably nonsense in real life, but it's in an NBME question somewhere. - Does not occur centrally on NBME exam, unlike small cell and squamous, and hence will be described as apical or peripheral lung lesion in non-smokers. - Apical tumors can are known as Pancoast tumors and cause Horner syndrome (miosis, ptosis, anhidrosis) due to impingement on C8 superior cervical ganglia (sympathetic nerves). They can also cause SVC syndrome (flushing of the face + congestion of neck veins) or brachiocephalic syndrome (only right side of face/neck) due to impingement on venous return. (+) Pemberton sign is worsening of flushing + neck vein congestion when raising the arms above the head. - Can be associated with migratory thrombophlebitis (Trousseau sign of malignancy). The latter is not limited to head of pancreas cancer. Adenocarcinomas in general are known to be associated with hypercoagulable state due to malignancy. USMLE won't ask specific mechanism, but liberation of tissue factor (factor III) by these cancers is a proposed etiology. - Associated with hypertrophic osteoarthropathy (clubbing + hand pain due to lung cancer); mechanism is fibrovascular proliferation. Literature says adenocarcinoma of lung is most common cause, although the association isn't 100% specific.
Large cell carcinoma	- Nonexistent lung cancer on USMLE. Bogus/garbage diagnosis. I don't think I've ever seen this assessed.
Bronchogenic carcinoid tumor	- Pulmonary nodule that secretes serotonin or serotonin-like derivatives, resulting in carcinoid syndrome (i.e., tachycardia, flushing, diarrhea). - A type of neuroendocrine tumor. - Carcinoid tumors are classically appendiceal and of the small bowel, but you should be aware that bronchogenic carcinoids exist. - USMLE wants urinary 5-hydroxy indole acetic acid (5-HIAA) for initial step in Dx.
Mesothelioma	- Cancer of "mesothelial cells" (answer on NBME) seen in patients with prior occupational exposure to asbestos. - Asbestosis occurs first, which then gives rise to mesothelioma years later. - Shipyard and construction workers are buzzy for prior occupational exposure. - Asbestosis will be described as pleural or supradiaphragmatic plaques ("soft tissue plaques seen on CXR"). Pulmonary biopsy shows ferruginous bodies. - Calretinin (+); a protein that is highly indicative of mesothelioma on staining. - Mesothelioma appears as a whiteish cancer and is described as an "encasing rind of pleural-based tumor" (i.e., circumferentially surrounds/wraps around the lungs). 

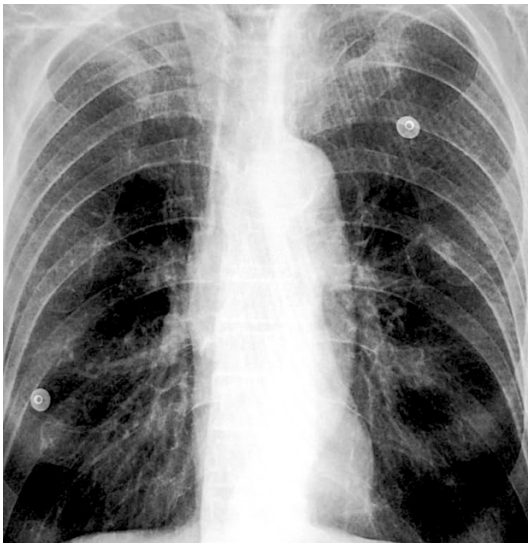
Nasopharyngeal carcinoma	- Can be caused by EBV. - A type of squamous cell carcinoma of the airway.
Laryngeal cancer	- Squamous cell carcinoma of vocal cords. - Smoking is major risk factor. - New NBME Q wants you to know this spreads to **cervical lymph nodes.**
Laryngeal papillomatosis	- Pediatric condition characterized by warts of the vocal cords. - Due to HPV 6/11 exposure from maternal vaginal canal. - Lesions will have papillary structures on biopsy. 

Pneumoconioses for USMLE	
Asbestosis	- As already mentioned above in the mesothelioma section, this is associated with shipyard workers, construction workers, and electricians. It can give rise to mesothelioma later in life.  - The above ferruginous body is dumbbell-shape. Choose macrophage as the answer on USMLE for the cell that initiates pulmonary fibrosis (in response to asbestos and in general). - Causes restrictive lung pattern.
Berylliosis	- Occupational exposure to beryllium in the aerospace / aeronautical industry. - Causes restrictive lung pattern. - Can cause granulomas.
Silicosis	- Occupational exposure to silicon (sand) in foundry or stone quarry workers. - Can cause egg-shelf calcifications in upper lobes. - Increases risk of tuberculosis infections. - Avoid anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) in these patients due to increased TB risk (TNF-α needed to suppress/fight TB).
Anthracosis	- Aka “coal miner’s lung.” - Black discoloration of the lung. - Can be either obstructive or restrictive.
Caplan syndrome	- Rheumatoid arthritis + any pneumonociosis, presenting as pulmonary nodules. - Clinical relevance is that patients with RA are at increased risk for developing pneumoconioses if they have a workplace exposure.

<table>
<tr><th colspan="2">HY General lung conditions for USMLE</th></tr>
<tr><td>Idiopathic pulmonary fibrosis
(Usual interstitial pneumonitis)</td><td>- HY DDx on USMLE for a patient over the age of 50 who has 6-12+ months of unexplained dry cough. This is how it shows up 4/5 times.
- Textbook restrictive lung disease, with ↔ or ↑ FEV1/FVC. The reason for the FEV1/FVC being greater than in obstructive lung disease is radial traction, as mentioned earlier.
- CXR and CT scan show “reticular” or “reticulonodular” pattern. These descriptors are exceedingly HY on USMLE, where students will overlook them in the vignette, but they are hugely buzzy for restrictive lung disease. They are colloquially known as “honeycombing,” but I have not seen the USMLE give a fuck about the latter colloquialism. They frequently just say “reticular” and “reticulonodular,” and then you know right away, “Boom. Restrictive lung disease,” i.e., fibrosis, etc. Tangentially, it’s to my observation that NBME will say “reticulogranular” frequently for NRDS, but the two vignettes are clearly disparate anyway.
 

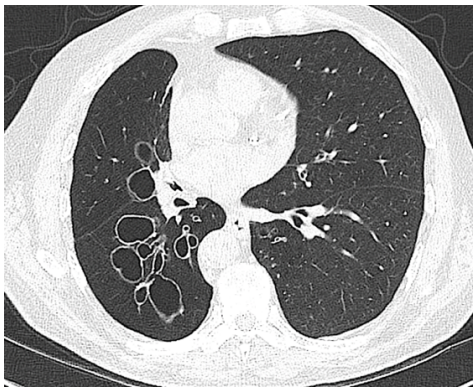
- After the CXR and spirometry are performed, 2CK wants “high-resolution CT of chest” as answer for next best step.
 
“Honeycombing” = reticular / reticulonodular pattern.

- New 2CK NBME wants “lung biopsy” as answer to confirm diagnosis of interstitial lung disease (i.e., idiopathic pulmonary fibrosis) after imaging.
- Vignette can also mention loud P2 (means pulmonary hypertension) with “dry inspiratory crackles heard bilaterally.”
- 1/5 times, the Q will be patient over 50 with increasing fatigue and shortness of breath over 6-12 months, with only 1 month of cough, where it initially sounds like heart failure, and they’ll say CXR shows “interstitial markings” instead of reticular/reticulonodular patterning.</td></tr>
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	However, they say patient has "↑ FEV1/FVC showing restrictive pattern" in the stem, which gives it away. - You need to know that **"usual interstitial pneumonitis" (UIP)** is another name for idiopathic pulmonary fibrosis. Yes, the name is weird, but it's not my opinion and it's asked twice on the NBMEs, where instead of writing "idiopathic pulmonary fibrosis" as the answer, they write "usual interstitial pneumonitis." UIP is technically a broad term that can refer to many restrictive lung conditions, but as I said, on NBME they use this synonymously with idiopathic pulmonary fibrosis. - **Tx on 2CK = pirfenidone** → anti-fibrotic agent that inhibits TGF-β-mediated synthesis of collagen. Another NBME also has simple "prednisone" as answer for Tx. So know both.
Chronic obstructive pulmonary disease (COPD)	- COPD = chronic bronchitis + emphysema. - Smokers will have combination of the two. When we say a smoker has COPD, we are saying he/she has chronic bronchitis + emphysema at the same time. - The term COPD can in theory apply to any obstructive disease of the lung that is chronic (e.g., asthma, Kartagener, etc.) but when the term is used without any specific condition attached, it refers to the combo of chronic bronchitis and emphysema.  - Hyperinflated lungs in COPD (due to air trapping) can push the heart to the midline. NBME will say there's a "long, narrow cardiac silhouette," or a "point of maximal impulse palpated in the sub-xiphoid space." In left ventricular hypertrophy, in contrast, there will be a lateralized apex beat, or a point of maximal impulse in the anterior axillary line. - Patients with COPD are chronic CO2 retainers, so they will have a chronic respiratory acidosis (i.e., ↑ CO2 and ↑ bicarb; pH can either be ↓ or compensated back into normal range). - "Exacerbation of COPD" as a diagnosis on USMLE will always give a patient with *high CO2*. This is really important. In other words, if they give you a big paragraph and you're not sure of the Dx, if you see CO2 is not elevated, the answer is not COPD exacerbation. - COPD exacerbations can be triggered by minor chest infections, so always **give antibiotics** on 2CK, even though etiology is usually viral. - First-line Tx is now considered to be a long-acting muscarinic receptor antagonist (i.e., LAMA such as tiotropium) or a long-acting β2 agonist (i.e., LABA such as olodaterol), either alone or in combination. - If insufficient, add an inhaled corticosteroid (e.g., fluticasone).

	- Should be noted that a question on an earlier Family Med form has ipratropium (SAMA) as the answer for 1st-line in COPD, but there aren't any other anti-muscarinic or β2 agonists listed. So the point is that if you are forced to choose a SAMA or SABA (i.e., albuterol) on USMLE, these are OK, but newer guidelines say start with a LAMA or LABA. - Home oxygen is indicated on 2CK if O2 sats are: - <88% saturation (55 mm Hg), or - <89% saturation (60 mm Hg) if the patient has cor pulmonale. - For 2CK, lung cancer screening with annual low-dose chest CT is done in patients who meet all of the following: 1) age 50-80; 2) have 20-pack-year Hx of smoking; and 3) smoked within the past 15 years.
Chronic bronchitis	- Chronic bronchitis = productive cough for at least 3 months in a year, for 2+ years. - Reid index >0.5 (ratio of the thickness of bronchial mucous-secreting glands to the bronchial wall itself). <0.4 is considered normal. - Chronic bronchitis is known as "blue bloater" because the mucous sits in the alveolar spaces and impairs gas exchange. This results in a shitload of hypoxic vasoconstriction → patient can become acutely blue and hypoxic during exacerbations. - The hypoxic vasoconstriction of pulmonary vessels causes pulmonary hypertension (i.e., if the vessels constrict, then pressure in the more proximal pulmonary arterioles increases). This increased afterload on the right heart can lead to right ventricular hypertrophy and right heart decompensation. When right heart failure (i.e., evidence of JVD or peripheral edema) occurs due to a pulmonary cause, we now call that cor pulmonale, as discussed in the Cardio PDF. - The pulmonary hypertension can cause a loud P2 and tricuspid regurg prior to cor pulmonale occurring. - Don't confuse chronic bronchitis with acute bronchitis, which can present as a worsening cough in patient (with or without COPD) following a viral infection. This is a temporary irritation/inflammation of the airways that is self-resolving.
Emphysema	- Emphysema = loss/destruction of alveolar surface area. - When alveolar surface area is reduced, so is alveolar capillary surface area, since the capillaries are within the alveolar walls → ↓ gas exchange. - Loss of capillary surface area causes pulmonary hypertension and ↑ afterload on the right heart, leading to cor pulmonale. This is because the capillaries can be thought of as transistors in a parallel circuit, where reducing the number of components in the circuit ↑ total resistance. - Known as "pink puffer," since although gas exchange is impaired, sudden and acute increases in hypoxic vasoconstriction, as with chronic bronchitis, are not a feature. - "Bullous changes" on CXR are synonymous with emphysema on USMLE. - Smokers can get centri-acinar emphysema (proximal alveolar structure is destroyed). - α1-antitrypsin deficiency patients get pan-acinar emphysema (entire alveolus destroyed).
α1-antitrypsin deficiency	- Codominant genetic condition resulting in emphysema and hepatic cirrhosis. - *ZZ* allele combo is worst and results in disease (asked on USMLE). - α1-antitrypsin is an enzyme produced by the liver that travels to the lungs and breaks down neutrophilic elastase. Elastase is an enzyme that normally causes damage to the alveoli. Homeostatically, elastase is required in small amounts for normal pulmonary function and cell

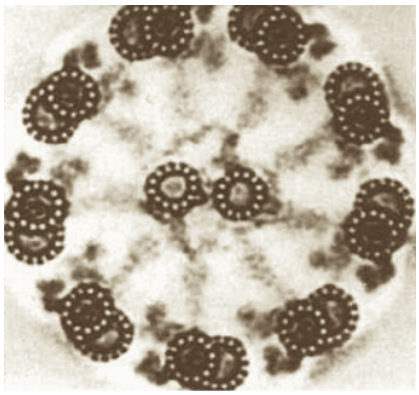
	turnover, but in high amounts it destroys the alveoli, resulting in emphysema. - The USMLE can have as an answer merely "deficiency of anti-protease." - Vignette will give young adult who has sibling or parent who's had early-onset emphysema or cirrhosis. - The USMLE Q can absolutely say the patient is a smoker or drinks alcohol. For instance, they might say the patient is 34 and has been smoking for 5 years and has bullous changes on CXR, or that the father died from alcoholic cirrhosis, and the student thinks, "Oh that can't be α1-antitrypsin deficiency though because they said smoking/alcohol." But the key point is that this condition *increases the risk* for early-onset emphysema and cirrhosis (i.e., it accelerates them). Normally, COPD should take 20-30+ years of smoking to develop, not 5.
Asthma	- Bronchospasm that occurs either idiopathically/hereditarily, or in response to certain allergens or cold air. - One-third of patients with asthma only present with a dry cough and no problems breathing. This is called cough-variant asthma. It will often present as a patient with a dry cough that's worse in the winter. - Can also present as part of atopy constellation – i.e., dry cough in winter, seasonal allergies / rhinoconjunctivitis / urticaria in spring, and eczema in summer. - USMLE wants you to know leukotrienes cause bronchoconstriction / asthma. They can give simple asthma Q + ask which of the following is most likely involved in the pathogenesis of this patient's condition, and the answer is just "leukotrienes," or "LTC4," "LTD4," or "LTE4." - Aspirin-induced asthma mechanism HY for USMLE → inhibition of COX by aspirin → shunting of arachidonic acid down lipoxygenase pathway → increased leukotrienes → increased bronchoconstriction. - Samter triad = aspirin allergy, asthma (due to aspirin), nasal polyps. - "Increased expiratory phase" is a buzzy phrase that will be thrown into quite a few asthma vignettes. It is not specific for asthma and can refer to any obstructive pathology, but I just make note of it here because you'll see it quite a bit for asthma and say, "What's that mean?" → in obstructive conditions, it takes a lot longer to exhale (FEV1 is ↓↓). - Acutely, asthma causes ↓ CO2, ↑ pH, ↔ bicarb. This is an acute respiratory alkalosis. Even though the patient is having difficulty breathing, CO2 diffuses quickly, whereas O2 diffuses slowly, so insofar as the patient's respiratory rate is ↑, CO2 will be ↓. Bicarb is unchanged because it takes the kidney about a day to alter excretion. In contrast, CO2 is ↑ in COPD because there is large amounts of hypoxic vasoconstriction due to excessive mucous (chronic bronchitis) or ↓ alveolar surface area, so CO2 can't get out, even with faster breathing. In asthma, alveolar surface area is intact, and the degree of mucous production and hypoxic vasoconstriction is not nearly as bad as chronic bronchitis. - The combination of ↓ CO2 and ↓ O2 in acute asthma attack is known as type I respiratory failure. Eventually, the patient will begin to fatigue and breathing rate will slow. This will be observed as CO2 and pH rebounding to normal levels *despite* O2 remaining low. This means the patient is in transition to a type II respiratory failure (i.e., hypoventilation), where we have ↑ CO2 and ↓ O2. USMLE wants intubation as the intervention. - 2CK Q gives vignette of asthma and then asks for best initial step in diagnosis → answer = "spirometry." The expiratory component of the curve, as discussed earlier, will appear concave. - Should be aware methacholine challenge can also be done to diagnose. This is a muscarinic agonist that bronchoconstricts and can induce

	symptoms. Never give during acute episode. This can be tried between episodes. But if you're forced to choose diagnostic modality, go with simple spirometry. - USMLE cares about both outpatient management of asthma as well as acute attacks. - For outpatient management: 1) First-line Tx is β2-agonist (albuterol) for acute attacks. 2) If the Q says the patient has weekly episodes, or they ask for what could decrease risk of future episodes if patient is already on albuterol, the answer is inhaled corticosteroid (ICS; i.e., fluticasone). 3) If the combo of albuterol + ICS is insufficient, the next step is increasing the dose of the ICS. 4) If that doesn't work, adding a LABA (i.e., salmeterol) is the next step. 5) If insufficient, agents such as mast cell stabilizers, leukotriene blockers, etc., can be tried. If patient has Hx of aspirin allergy, the latter in particular can be effective. 6) Last resort for outpatient asthma is oral prednisone. It is most effective at decreasing recurrence of episodes, but we want to avoid it if at all possible because of the risk of Cushing syndrome and ↓ linear bone growth in Peds. - Acute management of severe asthma attack (i.e., not outpatient) is nebulized albuterol, oxygen, and IV methylprednisolone. Inhaled corticosteroids have no role in acute asthma management. - NBME for 2CK wants you to know that any asthma patient who requires hospital management for an acute episode must automatically be given ICS (i.e., fluticasone) on discharge. In other words, if patient isn't currently on an ICS, it must be commenced at discharge. I'm explicit here because this particular NBME Q doesn't mention albuterol anywhere in the stem, but inhaled fluticasone is the answer. - Another 2CK NBME Q gives a patient who is currently not receiving any asthma medications but who gets 2+ episodes weekly → answer = "inhaled fluticasone + albuterol." Students think this is weird because they jump straight to dual management + are even audacious enough to put the ICS before albuterol in the wording of the answer, but it's what they want. Any patient with 2+ episodes per week needs ICS in addition to the albuterol and can be commenced right away on dual therapy.
Bronchiectasis	- Dilation of the airways (ectasia means dilation) that occurs due to "loss of musculature of the airways." - Transverse CT scan will show cystic dilation of the airways. 

	- Most common cause worldwide is TB. Most common cause in western countries is CF. But on USMLE, it will usually be a smoker. Asthma does not cause bronchiectasis. - Presents almost always as "cups and cups of foul-smelling sputum." - "Foul-smelling" means anaerobes, such as *Bacteroides*. - One question on 2CK Peds form gives bronchiectasis as the answer in a child who has right middle lobe syndrome, where they say there is scant white sputum (not cups and cups) and a thin, linear opacity visualized in the right middle lobe. If you think it's weird, complain to NBME not me. - **Clubbing** tends to be seen in these patients. It's not mandatory, but it's to my observation USMLE likes it for bronchiectasis.
Atelectasis	- Confusing word that refers to "lung collapse," or "collapse of alveoli." - Highest yield point for USMLE is that it is the most common cause of fever within 24 hours of post-surgery. If this is the first time you're reading this, that might sound weird, but this is pass-level and extremely important for 2CK. - There is one 2CK Q I've seen where they say a woman had a C-section two days ago and the answer was still atelectasis, so even though it's most common <24 hours, just be aware one Q exists where, oh em gee, it's 2 days later. - The mechanism is related to combo of pain meds + sedentation, where breathing becomes slower + shallower in hospital bed, leading to mild collapsing of some alveoli. This is why breathing exercises can be important post-surgery. - Will often present as **bibasilar shadows or opacities.** In other words, patient had surgery yesterday + now has fever + CXR shows mild opacity at the lung bases → answer = atelectasis. - NBME assesses obstructive (aka resorptive) atelectasis. This is when an area of lung distal to an obstruction from, e.g., a tumor, can cause alveoli to collapse. This then increases the chance for pneumonia distal to the obstruction. - 2CK IM form has "endobronchial obstruction" as answer for distal area of lung collapse (i.e., atelectasis) in patient with lung cancer; "vascular occlusion by tumor" is wrong answer (makes sense, as the tumor obstructs the respiratory tree, not blood vessel, in this case, but I've seen students accidentally choose the latter).
Obstructive sleep apnea (OSA)	- Can be obstructive (i.e., usually from obesity) or central (i.e., brain-related). What USMLE wants you to know: - These patients develop chronic respiratory acidosis – i.e., ↑ CO2, ↑ bicarb, pH ↔/↓. - Cor pulmonale can occur as a result of pulmonary hypertension from hypoxic vasoconstriction. JVD or peripheral edema will be seen with cor pulmonale. Descriptors such as RBBB, right-axis deviation on ECG, and wide splitting of S2 all mean RVH. If the patient merely has pulmonary hypertension but not yet cor pulmonale, the vignette can say loud P2 or tricuspid regurg (holosystolic murmur that ↑ with inspiration). - Chronic fatigue and poor oxygenation can lead to dysthymia / depression. The answer on NBME is "mood disorder due to a medical condition." - Polysomnography (sleep study) is what USMLE wants to diagnose.
Anaphylaxis	- Type I hypersensitivity in which IgE crosslinks on surface of mast cells and basophils, leading to degranulation and histamine + prostaglandin release. Eosinophils can be recruited in response. - Acute dyspnea and bilateral wheezes in patient who was gardening (stung by a bee), who ate a particular food (e.g., peanuts), or who was commenced on a recent drug (e.g., TMP/SMX).

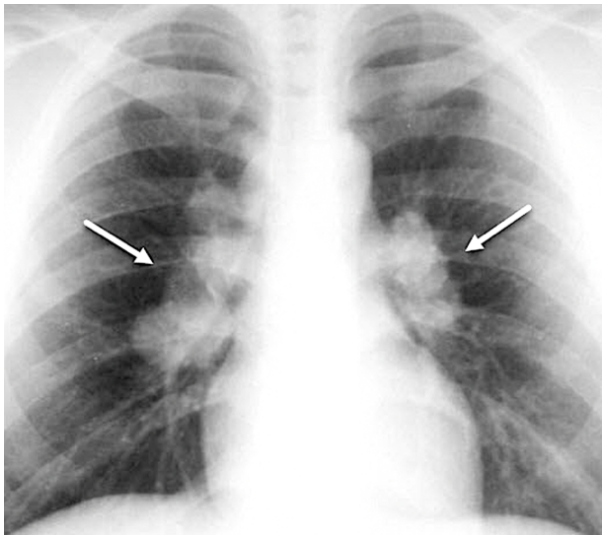
	- Vignette often gives tachycardia and low BP. - USMLE wants: ↓ vascular resistance, ↑ CO, ↓/↔ PCWP. - Tx = **intramuscular epinephrine** → the strong β2-agonistic effect opens the airways; the strong α1-agonistic effect constricts the arterioles and restores BP. - 2CK NBME asks, "In addition to self-injectable epinephrine therapy, what is most appropriate therapy to ↓ recurrences" → answer = **venom immunotherapy (VIT).** What this does is desensitizes the patient to the antigen by allowing him/her to develop neutralizing IgG antibodies against it.
Scombroid	- Often misdiagnosed as seafood allergy. - Bacteria in decaying meaty fish (e.g., mahimahi) can break down histidine in the fish into histamine, which causes an allergic-/asthma-like presentation. - Patients with history of asthma have greater risk of mortality.
Shellfish allergy	- Can cause allergic-/asthma-like presentation. - Occurs literally after eating shellfish (i.e., clams, mussels). - Don't confuse with scombroid. Students get trigger-happy when they learn weird conditions. If USMLE gives shellfish, the answer is shellfish allergy, not scombroid. The latter is from meaty fish and not an allergy.

HY Congenital lung diseases	
Cystic fibrosis	- Autosomal recessive; chromosome 7. - Mutations in *CFTR* gene, which codes for a chloride channel. - Most common mutation is ΔF508, which is deletion of a phenylalanine. - USMLE wants "abnormal protein structure," or "abnormal protein folding," a common mechanistic answer for CF. - In the disease, the CFTR chloride channel, normally located at the cell surface, is instead sequestered at the RER. This is a HY point, where they want you to know patients with disease do *not* have the chloride channel on the cell membrane. - Disease results in impaired ability to secrete Cl^- into alveolar and pancreatic secretions. This causes greater (-) charge within the cell. Na^+ then moves through ↑ ENaC into the cell to balance charge. Water follows Na^+, resulting in dried up alveolar and pancreatic secretions. These are known as inspissated secretions. Inspissation means desiccated (dried up) within a lumen. The inspissated alveolar secretions lead to mucous plugging, airway obstruction, and recurrent pneumonias. The inspissated pancreatic secretions lead to "exocrine pancreatic insufficiency" (HY phrase) and fat-soluble vitamin malabsorption. It is to my observation that vitamin E deficiency is the highest yield vitamin deficiency due to CF across NBME questions, which will present as neuropathy in a child who has CF. - Meconium ileus HY; refers to failure to pass stool in the first 24 hours after birth. - CF can sometimes cause nasal polyps. - USMLE doesn't obsess over pathogen causes for CF pneumonia, but classically *Pseudomonas* is a culprit. Prior to age 10, *S. aureus* exceeds *Pseudomonas* as most likely organism; after age 10, *Pseudomonas* eclipses *S. aureus*. The caveat is you have to use your head though: if they say 8-year-old with pneumonia due to gram-negative rods, you know that's *Pseudomonas*. Likewise, if they say a 17-year-old with pneumonia due to gram-positive cocci in clusters, you know that's *Staph aureus*.

	- Sweat glands reabsorb Cl^- rather than secrete it. This is impaired in CF, so we have ↑ Cl^- in the sweat (i.e., >60 mEq/L). The sweat chloride test is more accurate for diagnosis than genotyping due to allelic heterogeneity of *CFTR* gene, where many different mutations cause the disease and can be difficult to detect with routine screening panels. - A transepithelial nasal voltage test can also be done, where there is ↑ nasal potential difference. - Male patients can have congenital bilateral absence of vas deferens (CBAVD), leading to absent sperm in a sample. - Since this is an AR condition, there is a 2/3 chance a phenotypically normal sibling of a CF patient is a carrier. This applies to any AR disorder, but USMLE likes to use CF as the archetypal example. If this 2/3-point sounds confusing, I talk about this stuff in detail in my HY Genetics PDF. - CF medications not HY for Step, but you could be aware that -caftor agents (e.g., ivacaftor, lumacaftor) facilitate with proper CFTR channel localization to the cell membrane + structural folding. - Dornase-alfa (correct, not alpha) is a nucleotidase that softens mucous. - Guaifenesin is also a mucous-softening agent.
Primary ciliary dyskinesia (Kartagener syndrome)	- Answer on USMLE for a question that sounds like CF but patient has situs inversus or dextrocardia – i.e., patient will have recurrent pneumonias and organs (or just heart) on opposite side of body. - Abnormality of cilia function due to defective dynein arm. This is a HY point, where sometimes the answer will just be "dynein." - A cilium on cross-section has a 9x2 microtubule configuration. Dynein is a molecule that is necessary for cilia function.  - Patient will sperm in sample, but motility will be decreased. This is in contrast to CF, where sperm are absent due to CBAVD. - Women can get ectopic pregnancy due to abnormal Fallopian tube cilia. - This diagnosis sounds straightforward enough, but I've seen plenty of students miss these questions. For the situs inversus, they will describe this as cardiac sounds loudest on the right, or there being a large mass palpable beneath the left costal margin (liver on the opposite side).
Primary pulmonary hypertension (PPH)	- Answer on USMLE for a woman 20s-30s, non-smoker, who has increased pulmonary vascularity/markings and either a loud P2 or triscupid regurg (these are HY findings for pulmonary hypertension). - "Primary" means the pulmonary hypertension inherently *starts with* the lungs and is not due to a secondary (i.e., external) cause such as smoking, CF, systemic sclerosis, etc. - Due to mutations in BMPR2 gene. - Do not confuse this with cor pulmonale. Recall that the latter is right heart failure findings or RV structural changes due to a pulmonary cause. If the question gives evidence of RVH (i.e., RBBB, right-axis deviation on ECG, boot-shaped heart on CXR), or has overt right heart failure findings

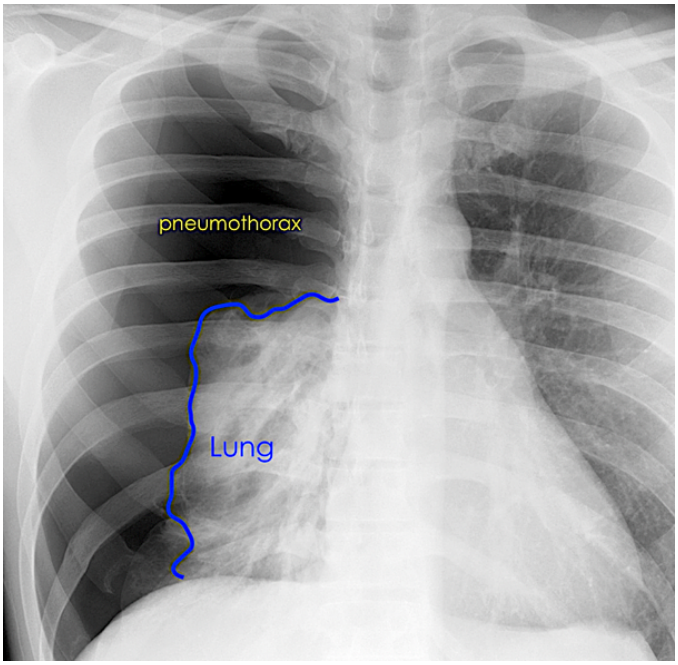
	(i.e., JVD or peripheral edema), then we can say the patient now has cor pulmonale due to primary pulmonary hypertension. A loud P2 and tricuspid regurg, however, are HY findings for pulmonary hypertension that don't necessarily mean cor pulmonale. - One of the highest yield points is that endothelin-1 is ↑ in these patients. This is a vasoconstrictor that is ↑ in pulmonary hypertension from any cause, but notably USMLE likes it for PPH. - Bosentan blocks endothelin-1 receptors and is HY Tx.

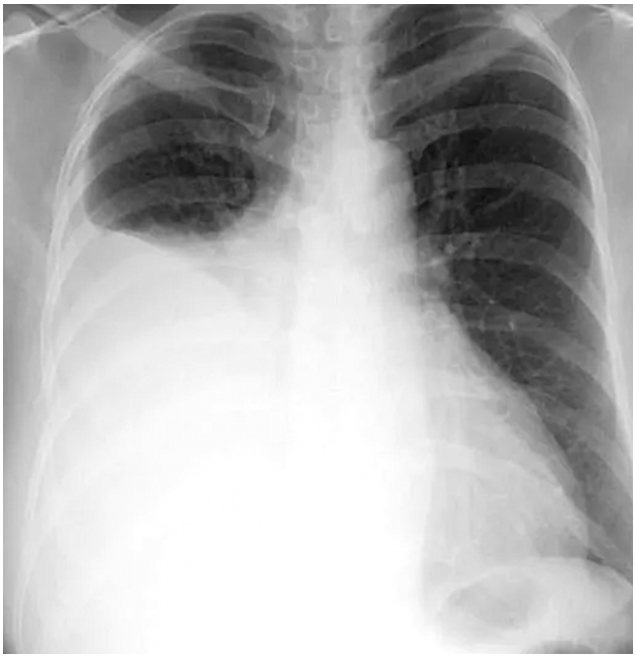
HY Autoimmune-related pulmonary conditions	
Systemic Sclerosis (aka scleroderma)	- Idiopathic autoimmune disease characterized by multi-organ system fibrosis and hardening of tissues (i.e., sclerosis). - Divided into limited and diffuse subtypes. - **Limited scleroderma** = **CREST** syndrome (**C**alcinosis, **R**aynaud's, **E**sophageal dysmotility, **S**clerodactyly, **T**elangiectasias). Associated with anti-centromere antibodies. - USMLE can describe Raynaud as color change of the fingers with cold weather. They describe sclerodactyly as tightening of the skin of the fingers. Esophageal dysmotility presents as GERD. I haven't seen NBMEs give a fuck about calcinosis, which is abnormal Ca^{2+} deposition in tissues. - **Diffuse scleroderma** = CREST syndrome + renal involvement (presenting as ultra-HY BP, e.g., 220/120, due to renal involvement causing a surge in RAAS). If patient doesn't have high BP, it's not diffuse type on USMLE. Associated with anti-topoisomerase antibodies (anti-Scl70). - Apart from just being able to diagnose these conditions, the highest yield point on USMLE is that both types cause **pulmonary fibrosis,** leading to **pulmonary hypertension.** - Pericardial fibrosis can also occur. - An offline Step 1 NBME wants ↓ LES sphincter tone and ↓ esophageal peristalsis as an answer. - USMLE wants "dress warmly in cold weather" as an answer for how to ↓ recurrence of Raynaud. Dihydropyridine CCBs can also be used (e.g., nifedipine), which ↑ arteriolar + capillary diameter. - USMLE wants to know which drugs you avoid in patients with Raynaud → answer = α1 agonists (e.g., phenylephrine, oxymetazoline), since these constrict arterioles/capillaries.
Sarcoidosis	- Idiopathic autoimmune disorder that is one of the highest yield conditions on USMLE. You must know this condition extremely well. - Characterized by non-caseating granulomas within lung tissue. These consist of activated macrophages called epithelioid macrophages, aka histiocytes. - These histiocytes in the lung secrete 1α-hydroxylase (normally produced in the PCT of the kidney in response to PTH), which will convert inactive 25-OH-D3 into active 1,25-(OH)2-D3 (i.e., causing high vitamin D; aka hypervitaminosis D), which then goes to the small bowel and ↑ absorption of calcium, causing **hypercalcemia.** - Even though 1,25-(OH)2-D3 also ↑ small bowel absorption of phosphate along with calcium, it's to my observation that phosphate can be *normal* in sarcoidosis NBME Qs, so do not be confused if you see phosphate in the normal range and only see hypercalcemia. - PTH is *suppressed* in sarcoidosis due to the high Ca^{2+} (exceedingly HY). - In other words, on USMLE you will select ↑ 1,25-(OH)2-D3 and ↓ PTH. - Archetypal presentation is African-American woman 20s-30s with 6+ months of dry cough and red shins (erythema nodosum). Other findings like low-grade fever with flares can occasionally be seen.

	- **Bihilar lymphadenopathy seen on CXR or CT.** They can also describe this as CXR or CT "shows hilar nodularity."  - There is a Family Med Q that gives early-30s African-American woman with dry cough + they say CXR shows no abnormalities → answer = "activation of mast cells" (i.e., asthma is the answer); wrong answer is "non-caseating granulomatous inflammation." So your HY point here is: if the vignette sounds like it could be sarcoidosis but they say CXR is normal or shows "mild hyperinflation" (also buzzy for asthma), choose asthma over sarcoidosis. - A 2CK IM CMS form gives a sarcoidosis vignette where they mention bihilar lymphadenopathy on a CT scan + high hepatic AST + ALP + weight loss; they don't mention hypercalcemia. Unusual combo of findings, but the answer is inferable based on the CT. - Lupus pernio is an enlarged nose due to sarcoidosis (not SLE, despite the name). - USMLE wants steroids (i.e., oral prednisone) as treatment.
Rheumatoid lung	- Rheumatoid arthritis can cause a restrictive lung disease known as rheumatoid lung. - In addition, methotrexate, the first-line DMARD for RA, can cause pulmonary fibrosis. So patients with advanced RA can have pulmonary fibrosis from a combo of the two.

HY Pulmonary embolic conditions	
Pulmonary embolism	- DVT that has embolized up the venous circulation to the pulmonary arteries. - Presents as acute-onset shortness of breath and tachycardia in patient with one of the following risk factors: post-surgery (including C-section); sedentation, e.g., due to trauma such as hip injury; OCP use; and thrombotic disorders (i.e., FVL, prothrombin mutation, MTHFR). - S1Q3T3 is most specific finding on ECG for PE that is nonexistent on USMLE and will get you questions wrong. What you want to remember is: **sinus tachycardia** (i.e., normal ECG but HR is simply high) is the most common ECG finding for PE. - First step in management (2CK) is heparin. - After heparin, do a spiral CT of the chest to diagnose. - If the patient is already on warfarin and gets a PE, the first answer they want is CT to diagnose, followed by IVC filter. Do not choose IVC filter prior to confirming the PE with a CT. - One 2CK NBME Q has tPA as the answer for PE in patient who has obstructive shock (i.e., low BP due to PE). - V/Q scan is done instead of CT first-line in pregnancy.

	- One 2CK Q gives V/Q that is performed and shows "multiple sub-segmental perfusion defects" → answer = recurrent pulmonary emboli. - Most common cause of death from PE is ventricular fibrillation → acute right heart strain from a saddle embolus precipitates the Vfib. - The acid-base arrows USMLE wants for PE are: ↓ CO2, ↑ pH, ↔ bicarb. In other words, we have an acute respiratory alkalosis from high RR. Bicarb is unchanged because the kidney requires minimum 12 hours to accomplish this. These arrows are the same as acute asthma attack. - One 2CK Q gives lab values in PE of ↓ CO2 and ↓ pH. What this means is: patient has lactic acidosis (hence ↓ bicarb) due to ischemia from poor perfusion. It's HY for USMLE that you know any cause of shock, whether that be septic, hypovolemic, cardiogenic (or even obstructive from PE), can cause lactic acidosis and low bicarb. Don't be confused by this. - PE is textbook example of **dead space** (as discussed earlier).
Amniotic fluid embolism	- The answer on USMLE if they say 30 seconds to 2 minutes after delivery of the placenta, the patient gets shortness of breath and tachycardia → sounds like PE, but answer is amniotic fluid embolism to the lungs. In contrast, if they say a woman had a C-section 2 days ago + now gets up in hospital to use bathroom + now has acute-onset SoB and tachycardia → answer = PE, not amniotic fluid embolism. - Can cause disseminated intravascular coagulation (DIC). It's one of the weird causes for USMLE. They can say the patient has bleeding from IV sites / catheter lines following delivery of the placenta.
Fat embolism	- The answer on USMLE if they say long-bone fracture (i.e., usually femur) in patient who now develops shortness of breath + petechiae on the shoulders/chest. - The patient will also have thrombocytopenia, but only about half of vignettes mention this. - The petechiae on the shoulders/chest are not due to the ↓ platelets, since conditions like ITP, for instance, don't present with this finding. Literature suggests it might be related to microvascular plugging by fat in the dermis. - Question can mention pulmonary biopsy stained with Oil Red O or Sudan black (stains for fat).
Air embolism	- The answer on USMLE if they say patient has sudden death following insertion of a central venous line → air goes to the lungs + kills patient.
Cholesterol embolism	- Will present as two ways on USMLE: 1) violaceous lesions on the feet in someone who just had a AAA repair → cholesterol plaque launches off to feet + they show you image like this one below:  The second way (2CK NBME 10) has cholesterol emboli Q where they don't show an image but describe the violaceous foot lesions in someone with mere Hx of AAA (rather than undergoing repair). They say "Maltese crosses are seen." Obscure, but apparently refers to birefringence of cholesterol esters that can be visualized under polarized light.

HY Trauma and pleural space conditions	
Pulmonary contusion	- Difficult diagnosis on USMLE forms. - Diagnosis of exclusion, meaning all of these Qs require eliminating the other answer choices to get there. - The Q will say patient was in an accident + now has some form of difficulty breathing and patchy infiltrates in one or more areas of lung. They may or may not mention rib fractures just above the areas of infiltrates. Sounds vague as I said, but that's what USMLE will say. - Textbook descriptors such as "white-out of the lung" are nonsense. I don't think I've ever seen this on NBME material. - Contused (bruised) lung is known to be fluid-sensitive, where the Q can say patient was given IV saline and now has worsening of O2 sats. This is classic, but I've also seen this in myocardial contusion questions, making the Dx difficult as I said. But you'll be able to eliminate myocardial contusion if they don't mention severe bruising or pain over the sternum or post-MVA arrhythmia (e.g., premature atrial contractions).
Flail chest	- The answer on USMLE forms if they say a patient has paradoxical breathing (i.e., the chest wall moves outward with expiration and inward with inspiration).
Ruptured bronchus	- The answer on USMLE forms if they say there's a "persistent air leak despite placement of a chest tube." This will be in a trauma patient.
Diaphragmatic rupture	- The answer on USMLE forms for an MVA followed by CXR showing "obscured left hemidiaphragm and the NG tube present in the lower left side of the chest." - Don't confuse with ruptured bronchus above, or with esophageal rupture, which will tell you water-soluble contrast swallow is visualized in the mediastinum.
Pneumothorax	- Air in the pleural space. The terms almost always refers to a "spontaneous pneumothorax," which will be acute-onset sharp chest pain in a tall, lanky patient in teens or 20s due to "ruptured subapical bleb."  - Pneumothorax + low BP = tension pneumothorax. - Mechanism for tension pneumothorax is compression of the vena cavas leading to ↓ venous return → **can cause JVD.** - Simple/spontaneous pneumothoraces cause ipsilateral tracheal shift. - Tension pneumothoraces cause contralateral tracheal shift. - Tension pneumothorax need not be due to overt trauma. It will literally just be pneumothorax + low BP 9/10 times on USMLE. I've seen only one Q on a 2CK Surg

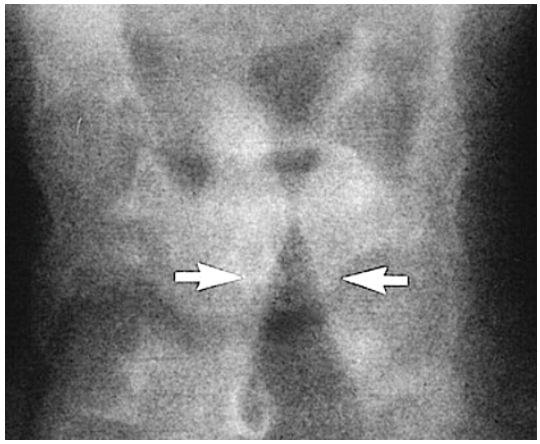
	form where BP is lower end of normal, but the patient is tachy around 120 bpm (implying barely holding BP in normal range). - Patient will have hyper-resonance to percussion, ↓ breath sounds, and ↓ tactile fremitus. Breath sounds are ↓ because the air in the pleural space masks the sounds of underlying air flow in the alveoli. Tactile fremitus is ↓ because the vibration of air in the alveoli is masked by the overlying air. - Treatment for pneumothorax is "needle decompression followed by chest tube." That phrase is HY. - One 2CK Q just has "tube thoracostomy" as the answer without needle decompression being listed. So if they don't list the needle, don't be confused, and just choose the chest tube. Thoracostomy means making a hole in the thorax. - Pleurodesis I've never seen as correct answer on NBME material but can be listed as distractor → means putting talc into pleural space to obliterate it in patients who have recurrent pneumothoraces. - Should be noted that very small pneumothoraces in stable patients with minimal symptoms can be observed. There is one Q on a 2CK NBME form where the answer is observe. But the vignette goes out of its way to emphasize how unremarkable the patient's presentation is. - There is a 2CK Q where pneumothorax is caused by barotrauma from ascending too quickly from underwater. This is different from Caisson disease ("the bends"), where nitrogen bubbles form in the blood. Pulmonary barotrauma from quick ascent, resulting in pneumothorax, can occur if some of the alveoli expand too quickly.
Pleural effusion	- Fluid in the pleural space; often refers to "hydrothorax," which means transudate or exudate of plasma-like fluid.  - USMLE wants: dullness to percussion, ↓ breath sounds, and ↓ tactile fremitus. Similar to pneumothorax, the latter two are ↓ because the fluid in the overlying pleural space masks air movement in the alveoli. - Many causes of pleural effusion on USMLE. I've observed left heart failure as a notable cause (i.e., patient will have pulmonary edema +/- pleural effusion). NBME can also give you pleural effusion with pneumonias, tuberculosis, and aortic dissection. - Meigs syndrome = triad of ovarian fibroma, ascites, right-sided pleural effusion.

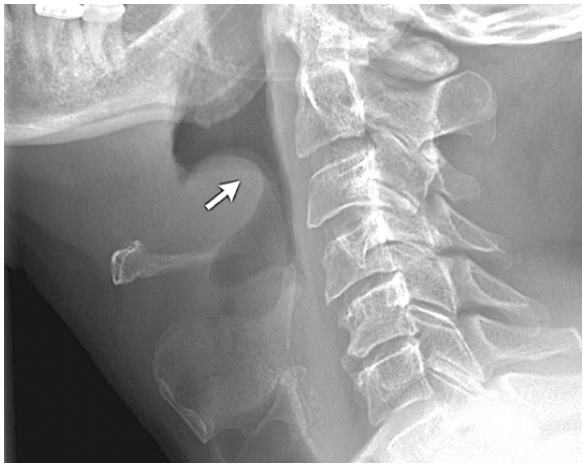
	- USMLE wants you to know transudative versus exudative pleural effusions → transudates are more water-like, with fewer solutes; exudates contain more solutes. As per Light's criteria (HY for both Steps 1 and 2CK), an exudate will have: - Pleural fluid protein : serum protein ratio >0.5. - Pleural fluid LDH : serum LDH ratio > 0.6. - Pleural fluid LDH >2/3 upper limit of normal of serum LDH. - In other words, transudate contains less protein and LDH as the two main distinctions. - Transudates contain fewer WBCs than exudates, but I routinely see 500 WBCs/μL in transudate Q on NBME. - Highest yield cause of transudate on USMLE is left heart failure, as discussed in the Cardio PDF. - ARDS, pulmonary emboli, and infections can cause exudative pleural effusions (due to inflammation). - Progression to empyema HY (see below).
Empyema	- Pus in the pleural space. - For 2CK Surg, it is exceedingly HY you know about empyema. There's two ways it presents on NBME: - 4 out of 5 questions, it will present as a sequence of pneumonia → parapneumonic effusions (i.e., exudative pleural effusions that occur due to pneumonia) → empyema. In other words, if a patient has a pneumonia that then leads to an exudative pleural effusion, we simply call that a parapneumonic effusion. This effusion can become progressively worse and purulent, to the point that the pH of the pleural fluid falls below ~7.1ish. We now call it empyema, which is frank pus in the pleural space. **Low pH of the pleural fluid to < ~7.1** is very HY on Surg forms. I've seen Qs where they say 7.12, 7.11, and 6.99, and they want empyema. - 1 out of 5 Qs will not mention anything about Hx of pneumonia. They will give patient who has **persistent fever + recurrent pleural effusions** in someone who has cirrhosis (i.e., ↓ oncotic pressure → transudative pleural effusions). Then the answer is just empyema. - NBME wants tube thoracostomy to drain the purulence.
Chylothorax	- Chyle (lymph fluid) in the pleural space. - The answer when they tell you a patient has milky or white-cloudy fluid in the pleural space **where the pH is significantly above 7.1.** - For instance, they might give you 15-line paragraph where they say milky fluid + tons of lab values, but you notice the pH of the pleural fluid is 7.40. This means it's chylothorax and empyema is wrong. - Can be caused **iatrogenically by insertion of central line via the left internal jugular vein,** or by cancers, e.g., lymphoma.
Hemothorax	- Blood in the pleural space. - Apart from massive trauma, the way this shows up on USMLE is as **malignant pleural effusion.** - They'll give you massive paragraph with tons of info + tell you patient has history of breast or lung cancer + has dullness to percussion → answer = malignant pleural effusion. - Patient can have hypotension and muffled heart sounds, similar to cardiac tamponade, but rather than JVD (to complete Beck triad), can have **flattened** neck veins (due to loss blood from the circulation).

HY Alveolar fluid conditions	
Pulmonary edema	- As discussed in the Cardio PDF, this will be due to left heart failure on USMLE. - There is such thing as "non-cardiac pulmonary edema" (i.e., ARDS, TRALI [discussed below in this table]), but for USMLE purposes, if you see "pulmonary edema" as an isolated phrase, it refers to transudation of fluid into the alveolar spaces due to ↑ pulmonary capillary hydrostatic pressure from left heart pathology. - PCWP and LAP are both elevated. - USMLE can give you vignette of, e.g., MI with dyspnea, and they ask for the mechanism of the dyspnea in the patient → answer = "increased alveolar-arteriolar oxygen gradient" – i.e., the patient can breathe just fine so alveolar O2 is normal, but fluid impairs the gas exchange, so we have low arterial/arteriolar oxygen. - "Cephalization of pulmonary vessels" is buzzy and synonymous with pulmonary edema on USMLE. Shows up in some NBME vignettes.
Acute respiratory distress syndrome (ARDS)	- The answer on USMLE for **bilateral** exudative chest infiltrates and ↓ O2 sats in patient following: pancreatitis; **aspiration of vomitus;** near-drowning episodes (aspiration of fresh/sea water); improper insertion of NG tube into the lungs with feeding initiated; toxic shock syndrome; or general trauma / sepsis. - NBME Q gives vignette of toxic shock syndrome and then asks most likely cause of death in this patient → answer = ARDS. - Pulmonary decompensation associated with pancreatitis is very buzzy. - Another Q gives patient who is brought to hospital following near-drowning episode + they ask what the patient needs to be monitored for → answer = ARDS. - ARDS is technically defined as a PaO2/FiO2 <300, but USMLE doesn't give a fuck. I've never once seen them apply this ratio or care. - I emphasize up above the bilateral nature of ARDS because many Qs give ARDS as DDx for unilateral conditions and it's wrong. - In theory, patient can be ventilated as follows: prone positioning (patient on stomach) + low-tidal volume setting + permissive hypercapnia. IIRC this is asked once on a 2CK form. - An offline Step 1 NBME Q wants "increased surfactant protein D" as the answer in a patient who is recovering from ARDS. Apparently surfactant protein D is a marker of lung injury and is ↑ in patients with ARDS (or who are recovering from it). Call it weird all you want but it's on the NBME.
Transfusion-associated lung injury (TRALI)	- The answer on USMLE if a patient has an ARDS-like presentation with bilateral crackles and low O2 sats **<6 hours** following a transfusion. - Mechanism is abnormal priming of neutrophils in the lung that react to cytokines within transfused blood products (this is straight from a CMS Surg form 8 explanation). - This is technically a type of non-cardiogenic pulmonary edema.
Transfusion-associated circulatory overload (TACO)	- Aka transfusion-induced hypervolemia. - Differs from TRALI in that this is a type of cardiogenic pulmonary edema (i.e., the left heart can't handle the ↑ hydrostatic pressure from ↑ volume, so transudation into the alveoli occurs). - Annoying diagnosis but presents two ways: 1) If a patient *with Hx of heart failure or MI* develops respiratory distress following transfusion of repeated blood products. - A 2CK NBME Q gives a 72-yr-old with Hx of MI ten years ago who gets shortness of breath and bilateral crackles 30 minutes after transfusion with crystalloid solution and 4 packs of RBCs (they don't specify the volume of crystalloid in the Q).

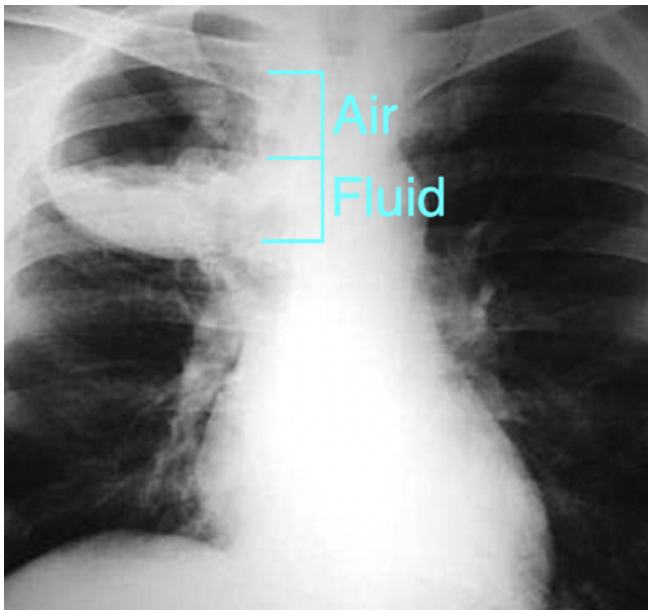
	2) They don't mention Hx of cardiovascular disease + the vignette will sound exactly like TRALI **> 6 hours** following a transfusion. - Surgery form 7 for 2CK gives an elderly dude who received only 3 packs of RBCs + he develops bilateral crackles and low O2 sats, but they say this occurs 12 hours after admission (not <6 hours as with TRALI), where the answer is "X-ray of the chest" as the next best step in diagnosis. The Q doesn't rely on you discerning TACO vs TRALI to get it right, but the explanation says it's TACO. In summary, if ARDS-like picture after transfusion: If <6 hours → TRALI. If >6 hours → TACO. If heart disease → TACO regardless of time frame.

HY Infection-related stuff	
Pneumonia	- Lobar pneumonia = *Strep pneumoniae* on USMLE (right lower lobe consolidation with dullness to percussion). - Bilateral interstitial pneumonia (aka atypical pneumonia) in immunocompetent patients = *Mycoplasma* on USMLE. - Lobar pneumonia where they say "interstitial markings" and *Strep pneumo* isn't listed → answer = *Mycoplasma* (the word "interstitial" wins over location). - Bilateral interstitial / "ground-glass" pneumonia in AIDS patient → *Pneumocystis jirovecii* pneumonia (PJP). - Lobar pneumonia in AIDS patient → *Strep pneumo*, not PJP. - Bacterial pneumonia specifically post-influenza infection → *S. aureus*. - Bilateral pneumonia + low Hb or (+) Coombs test → *Mycoplasma* → can cause cold agglutinins, which means IgM against RBCs → hemolysis. - Pneumonia + hyponatremia and/or diarrhea → *Legionella*. - Pneumonia + business conference or residential home (implies air conditioners) → *Legionella.* - Pneumonia in 3-wk-old neonate who had conjunctivitis 1-2 weeks ago → *Chlamydia trachomatis* (the STI; drains through nasolacrimal duct to lungs). - Pneumonia in newborn first few days of life → Group B Strep (*Strep agalactiae*), which is gram (+) cocci. If they say gram (+) rods, that's instead *Listeria*. If they say gram (-) rods, that's *E. coli*. - Pneumonia + rabbits → *Francisella*. - Pneumonia + bird keeper → *Chlamydia psittaci.* - Pneumonia + southwest US and/or earthquake dust → *Coccidioides*. - Patients who have lung cancer are prone to obstructive pneumonias (on 2CK form). - Pneumonia in CF → *Pseudomonas* or *S. aureus*. - Pneumonia in patient with central venous catheter + right upper lobe lesion → answer = *Staph epidermidis* (on NBME). Cather = biofilms. - USMLE wants for pneumonia: adventitious/bronchial (i.e., abnormal) breath sounds + ↑ **tactile fremitus** (air vibrates due to movement through infective consolidation within alveoli). - Community-acquired pneumonia (CAP) empiric Tx = azithromycin on 2CK (on NBME). This covers the atypicals (*Mycoplasma, Legionella, Chlamydia)* as well as *S. pneumo*. - If patient has been on antibiotics in the past 3 months or has severe lung disease, levofloxacin (respiratory fluoroquinolone) can be given first-line.

	- CAP that results in sepsis or septic shock → give ceftriaxone (if listed, choose cefotaxime for peds). - Nosocomial pneumonia (i.e., hospital- or ventilator-acquired) requires coverage for MRSA and *Pseudomonas*. USMLE wants vancomycin PLUS either ceftazidime (a 3rd-gen cephalosporin) or cefepime (a 4th-gen). - For fungal pneumonia, Tx = fluconazole. - For fungal pneumonia + fungemia (high fever, chills) → Amphotericin B. - NBME for 2CK wants you to know **sputum culture, followed by blood cultures** are done in all patients with pneumonia who are septic. What they do in this patient is give you patient who has sputum culture performed, then they ask what should be done next for diagnosis? → answer = blood culture. - If you get *Pneumocystis* pneumonia, jump straight to **bronchoalveolar lavage** as the answer. - If you get a patient who has CXR or CT showing cavitary lesions in the lungs filled with a mass (likely *Aspergillus* fungus ball), they want "open lung biopsy" as the most confirmatory test. - When a pneumonia fully resolves and USMLE asks why the CXR is normal after the fact, the answer is "maintenance of integrity of basement membranes," as mentioned earlier. - 2CK wants you to know that pneumonias can occur in patients distal to areas of lung obstructed by tumors; the answer will be "endobronchial obstruction" as the reason the patient has distal pneumonia (same answer for why there can be distal atelectasis). - Step 1 NBME wants you to know **rituximab** (monoclonal antibody against CD20 on B cells) increases the risk of bacterial pneumonia. This makes sense, since B cells are required for antibody production as part of humoral immunity against bacteria.
Bronchiolitis	- Answer on USMLE for a kid <18 months old who has low-grade fever and bilateral wheezes. - Caused by respiratory syncytial virus (RSV). - Tx is supportive care on USMLE. Don't choose answers like ribavirin or palivizumab.
Laryngotracheal bronchitis (aka croup)	- Caused by paramyxovirus (aka parainfluenza virus). - Presents as hoarse, barking, or seal-like cough in school-age kid. The Q can say the cough gets better when his dad brings him out into the cold air. - Neck x-ray shows "steeple sign," which is sub-glottic narrowing.  - Sometimes the Q can give you easy vignette of croup, but then the answer is just "larynx" (literally inflammation of the larynx, trachea, and the bronchi). "Sub-glottic" means below the area of the vocal cords. The larynx is the area encompassing the vocal cords.

	- Tx is supportive. If they force you to choose an actual Tx, the answer is steroids if listed, followed by nebulized racemic epinephrine.
Epiglottitis	- Caused by *Haemophilus influenzae* ***type B***. - Seen in unvaccinated and immigrants (can be unvaccinated), as well as patients with asplenia or sickle cell (auto-splenectomy). - Can also cause meningitis. - X-ray of neck shows "thumbprint sign."  - Presents as child who has fever + difficulty breathing. They can say the kid is drooling and/or in tripod positioning (facilitates use of accessory muscles). - USMLE wants intubation as answer. Epiglottitis is a medical emergency that can lead to sudden occlusion of the airway. - Tx = 3rd-gen cephalosporin (cefotaxime in peds, or ceftriaxone); give rifampin to close contacts.
Bacterial tracheitis	- Diagnosis of exclusion on USMLE (i.e., you eliminate to get there). - Classically caused by *S. aureus* following a viral URTI. - Shows up on 2CK form as patient with recent viral URTI who now has stridor (suggesting upper airway narrowing) + no drooling (suggests against epiglottitis) + does not improve with racemic epinephrine (suggests against croup) + **can fully open the mouth without difficulty.** - Being able to open the mouth without difficulty is important because this contrasts with peritonsillar abscess, where the patient has difficulty fully opening the mouth.
Pertussis	- Caused by *Bordetella pertussis*. - Classic whooping cough presents as succession of many coughs followed by an inspiratory stridor. - What you need to know for USMLE is that this can absolutely present in an adult and that they can be vague about it, just describing it as a regular cough. The way you'll know it's pertussis, however, is they will say there's either **hypoglycemia** or **post-tussive emesis,** which means vomiting after coughing episodes. - Pertussis can cause super-high WBC counts in the 30-50,000-range, where there are >80% lymphocytes (reactive lymphocytosis). *This makes it resemble ALL.* So you should know for Peds that ALL-like laboratory findings + cough = pertussis. - Q will ask number-one way to prevent → answer = vaccination (not hard, but they ask it). Pertussis is part of TDaP. The pertussis component is killed-acellular; the tetanus and diphtheria are toxoid. - Toxin inhibits G-alpha-I G-protein (leads to ↑ cAMP). - Erythromycin can be given to patients with active cough; USMLE doesn't give a fuck about pertussis stages.

	- Close contacts should also receive erythromycin.
Pleurodynia	- As mentioned in the Cardio PDF, this is an MSK condition I've seen asked twice on 2CK material that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. - Creatine kinase can be elevated in stem due to ↑ tone of muscle.
Viral pleurisy	- Viral infection causing inflammation of the pleural lining. - Presents as sharp chest pain. - Unlike pleurodynia, CK will not be elevated.
Tuberculosis	- *Mycobacterium tuberculosis* has unique cell wall composed of mycolic acid that is difficult to gram stain. Requires acid-fast stain. - Produces **cord factor** (asked on NBME) as a virulence factor. - Can present similar to lung cancer, where patient can have B symptoms (i.e., fever, night sweats, weight loss) and hemoptysis. - Living in a homeless shelter or immigrant status from endemic area is buzzy. I've seen rural India and Albania as two locations in NBME Qs. - Can cause cavitations and calcification in the lung grossly; on histo, causes caseating granulomatous inflammation. - Ghon foci/complexes are textbook descriptors for TB lesions but not assessed eponymously on USMLE. - Can cause constrictive pericarditis (can also calcify). - Disseminated TB (miliary TB) can affect multiple organ systems, leading to **psoas abscess, Pott disease** (TB infection of the vertebrae), adrenal insufficiency, meningitis, osteomyelitis, and arthritis. - First step in diagnosis is PPD test (type IV hypersensitivity). - If PPD test is (+), the next best step is CXR. - If PPD is (+) but CXR (-), next best step is "treat for latent TB, " or "give TB prophylaxis." This is isoniazid (INH) for 9 months + vitamin B6 (since INH can cause B6 deficiency). It is exceedingly HY you know that neuropathy in a patient being treated for TB has B6 deficiency. - If PPD and CXR are both (+), the next best step is "treat for active TB," which is RIPE for 2 months + RI for 4 more months (6 months total). RIPE = rifampin, isoniazid, pyrazinamide, ethambutol. - BCG vaccine is live-attenuated. USMLE wants you to know Hx of BCG vaccine does not change management based on PPD guidelines. - If USMLE asks you how long after TB exposure will someone's sputum cultures be positive, the answer is 2-5 weeks. - Interferon-gamma release assay (Quantiferon Gold) can be used in patients who have Hx of BCG to ↓ false (+)s, but USMLE doesn't assess it. The reason I mention it is because they want you to know *interferon-*γ is required for stimulation of alveolar macrophages to control TB. - Patients who have IFN-γ deficiency or IL-12 receptor deficiency (IL-12 normally ↑ IFN-γ) have ↑ susceptibility to TB infections. If this immuno stuff sounds confusing, I talk about this in detail in my HY Immuno PDF. - TNF-α is also required to suppress TB. Therefore drugs such as infliximab, adalimumab, and etanercept ↑ risk of TB, which is why they should be avoided in silicosis patients, as mentioned earlier. - What is considered a positive PPD test differs depending on risk factors: **- >5mm (+):** Hx of close contact to someone with active TB; immunocompromised patient (AIDS, organ transplant recipient receiving immunosuppressants, chronic corticosteroid user); calcification on CXR.

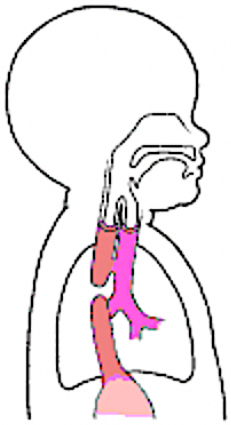
	- >10mm (+): Health care worker or prisoner/prison worker; immigrant from endemic area; TB laboratory personnel, children <4. **- >15mm (+):** everyone else. - If a PPD test is (+), never repeat it. If it is negative, it must be repeated in 1-2 weeks (i.e., sometimes false-negatives).
Hot tub lung	- Answer on a 2CK form for a guy who moves into a new apartment building with a hot tub and gets bilateral chest infection → inhalation of fumes with *Mycobacterium avium complex* (MAC).
Pulmonary abscess	- Exceedingly HY on USMLE. - USMLE wants "aspiration of oropharyngeal normal flora," or "aspiration of oropharyngeal anaerobes" as the cause. - Oropharyngeal normal flora = *Bacteroides* (strictly anaerobic gram-negative rods); as well as *Peptostreptococcus* and *Mobiluncus*. The latter two are not HY, but *Bacteroides* is. I mention all three, however, because the Q can say sputum sample shows "gram-negative rods, gram-positive cocci, and gram-positive rods," which refers to all three. But the bigger picture concept is, this = mixed normal flora. - The stem can say the patient has "foul-smelling sputum." This descriptor is exceedingly HY and is synonymous with anaerobes on USMLE. - Q will give aspiration risk factor, such as alcoholism, dementia (can cause loss of gag reflex), Hx of stroke (leading to dysphagia), or epilepsy. - Q can also mention broken or missing teeth (hypodontia) as risk factor. - Often described on NBME as pulmonary lesion with an **air-fluid level.** This is buzzy, but not a mandatory descriptor. This refers to the top half of the circle being air, and the bottom being pus, the latter settling due to gravity.  - **Tx = clindamycin**. USMLE loves this. Some students ask about ampicillin + sulbactam in relation to updated guidelines. I haven't seen this on NBME though. - If Q tells you patient was treated for pulmonary abscess + a year later there's still a lesion seen on CXR → answer = "failure of maintenance of basement membranes."
Acute bronchopulmonary aspergillosis (ABPA)	- Presents as asthma-like presentation in patient with ↑ sensitivity to aspergillus skin antigen.

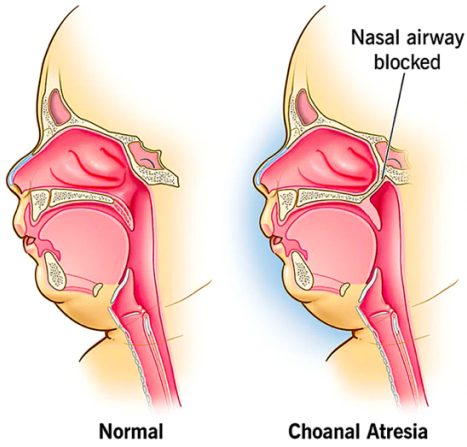
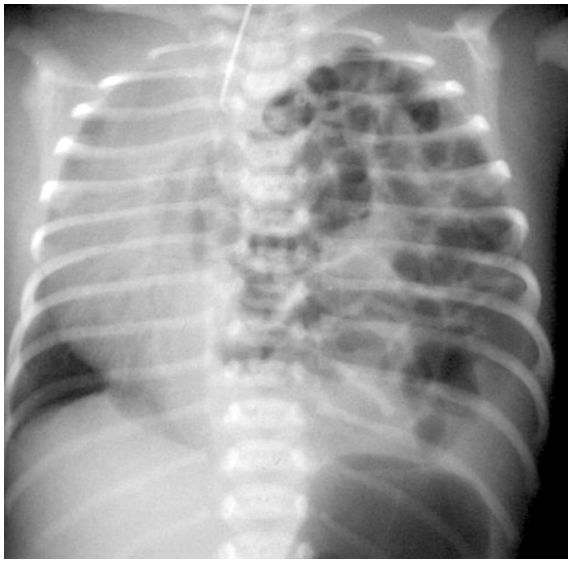
Hypersensitivity pneumonitis	- Answer on USMLE for bilateral lung condition + fever in farmer who has exposure to hay (on new Step 1 NBME). - They will tell you the fever self-resolves after 2 days and he now is afebrile. - Byssinosis (pneumoconiosis from hemp) is wrong answer, since this won't present with fever + classically presents in textile workers.
Cryptogenic organizing pneumonia (COP)	- Idiopathic restrictive lung disease where patient has pneumonia-like presentation that fails to improve with antibiotics. Not actual pneumonia. - Formerly known as bronchiolitis obliterans organizing pneumonia (BOOP). - Nonexistent yieldness on USMLE, but I mention it because you will sometimes see this as a *wrong* answer choice, particularly on hard 2CK Qs, and I've seen enough students erroneously pick it.
Otitis media (OM)	- Infection of portion of ear just deep to tympanic membrane. - Most commonly *Strep pnuemo*. - Will present as red, **immobile** tympanic membrane. Immobility of the tympanic membrane is highly sensitive for OM, meaning that if the Q says mobility is normal, we can rule out. - "Ear tugging" can be a sign in children of either otitis media or externa. - Tx is amoxicillin or penicillin. - Augmentin (amoxicillin/clavulanate) is classically given for *recurrent* OM. So if you are forced to choose between amoxicillin/penicillin alone or Augmentin, go with the former. - For 2CK Peds, a tympanostomy tube (aka grommet) is used if the kid has >3 OM occurrences in 6 months, or >4 in a year.
Serous otitis media	- Aka otitis media with effusion. - Presents as fluid behind the tympanic membrane in a kid weeks after resolution of 1 or 2 otitis media infections. - Almost always benign and self-resolves in 4-8 weeks. Answer is observation.
Tympanic membrane perforation	- "Tympanic membrane perforation" is the answer on new 2CK NBME for 2-year-old who had 3-day Hx of viral infection followed by awakening with severe ear pain + has dried blood on ear lobe and pillow + otoscopy cannot visualize tympanic membrane because of seropurulent fluid draining from the ear canal. - Can occur due to otitis media, although vignette on NBME doesn't sound like classic OM and is as described above.
Mastoiditis	- Inflammation of mastoid bone caused by untreated otitis media. - The mastoid process is the posterior part of the temporal bone that is felt just behind the ear. - Can present as a painful ear pinna that is displaced (e.g., upward and outward). - Diagnosis is made by **CT or MRI.** X-ray is wrong answer. - 2CK IM Q gives a 2-year-old with mastoiditis where the answer is "CT of the temporal bone." Sounds wrong, since this is ↑ radiation for a kid, but it's what they want.
Myringitis	- Isolated inflammation of the tympanic membrane. - Can be bullous (i.e., bullous myringitis). - Caused by *Strep pneumo* or *Mycoplasma*.
Otitis externa (OE)	- Infection of ear superficial to tympanic membrane. - Classically caused by *Pseudomonas*. - Increased risk in swimmers and diabetics. - An NBME form has "necrotizing otitis externa" as answer for black skin within the ear canal in a patient. This is aka "malignant otitis externa." - USMLE wants "acetic acid-alcohol drops" as prophylaxis in college student who does crew + continues to have water exposure. - Tx (not prophylaxis) = "topical ciprofloxacin-hydrocortisone" drops.
Sinusitis	- The answer if they tell you a school-age kid has a lingering fever after an upper respiratory tract infection (URTI) for 10-14+ days.

	- Whenever a URTI lingers for more than ~10ish days, you want to think about sinusitis as a differential. - A 2CK vignette gives nocturnal cough (reflects aspiration; in this case, from the sinuses) and grey membranes in the oropharynx. The grey oropharyngeal membranes detail sounds weird, since that is normally buzzy for *Diphtheria*, but it shows up on an NBME Q where the answer is sinusitis and *Diphtheria* isn't listed. - IgA deficiency Qs, which presents as recurrent sinopulmonary infections, can say patient has Hx of pneumonias + presents today with sore left cheek → reflects sinusitis. - For 2CK, **CT scan** is done if chronic sinusitis >12 weeks. After CT is performed for chronic sinusitis, nasal endoscopy can be performed. - For Step 1, they can say nasal endoscopy shows obstructed drainage from the maxillary sinus + ask for where a drainage cannula should be placed → answer = **"at the hiatus seminlunaris in the middle meatus."** If you think it sounds stupid or low-yield, take it up with the NBME exam. Student will then encounter it on exam and be like oh wow lol. - Tx is amoxicillin/clavulanate (Augmentin). This is in contrast to OM and *Strep* pharyngitis, which are treated with just amoxicillin or penicillin alone, without the clavulanate (unless recurrent).
Influenza	- Causes respiratory distress, fever, and myalgias (muscle pain). For USMLE purposes, the myalgias are exceedingly HY as a vignette finding that usually suggests the flu over other diagnoses. - Has 8 segments, two of which are hemagglutinin and neuraminidase. - Hemagglutinin mediates viral attachment to the cell by enabling its binding at sialic acid receptors. - If a question asks about the molecule most flu vaccines are targeted against, the answer is hemagglutinin. - Neuraminidase allows for newly synthesized viral particles to leave the host cell. This enzyme cleaves sialic acid residues, which normally bind the new viral particles within the cell. Once these residues are cleaved, the viral particles can leave the cell. - Drugs such as oseltamivir and zanamivir are sialic acid analogues that function as neuraminidase competitive inhibitors. In other words, they prevent the virus from leaving the cell. If the USMLE asks which drug prevents viral spread within a community, or they tell you a drug is given and now host cells are "packed with virions" (because they can't leave the cell), the answer is one of the -mivirs. - Antigenic drift is **point mutations** in hemagglutinin and/or neuraminidase, where the virus has changed slightly. It leads to seasonal epidemics. Antigenic shift is due to two influenza viruses entering a cell, one of human origin, the other of animal origin (such as bird or swine), where they engage in **reassortment of viral segments,** leading to a completely novel influenza virus. It leads to generational pandemics. - If a patient gets a bacterial lobar pneumonia following recent convalescence from influenza infection, USMLE likes *S. aureus* as a HY cause. The USMLE will not play trivia where they list *S. aureus* alongside *S. pneumo* and you're forced to choose. What they'll do is say something about how a guy recently recovered from a viral illness in which he had high fever and myalgias, and now he has a pneumonia caused by a gram-positive coccus in clusters --> answer = *S. aureus*. In contrast, *S. pneumo* is gram-positive diplococci.
Coronavirus	- The virus specifically known as SARS-CoV-2, or COVID-19, caused the 2019 global pandemic. - SARS stands for Severe Acute Respiratory Syndrome. - The pandemic is believed to have started following a laboratory leak in Wuhan, China, although this has been a source of political debate, where

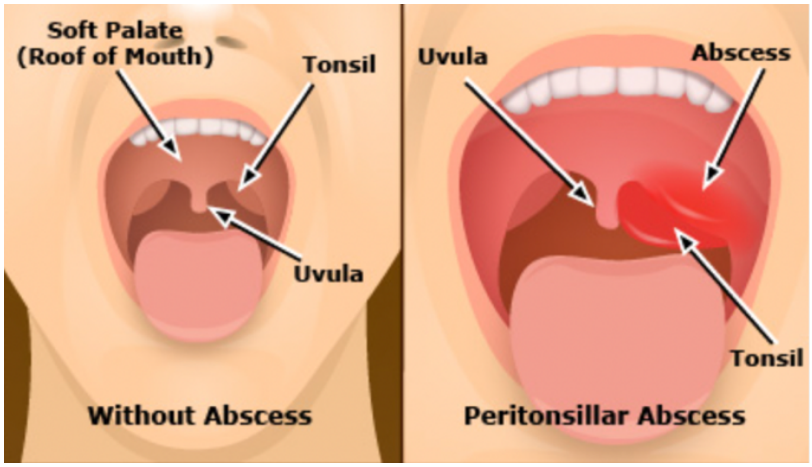
	initial explanations asserted that there was a natural, zoonotic origin for the virus (i.e., originating from animals, e.g., bats). - Has characteristic spike proteins that create a crown-like appearance on electron microscopy. - The spike proteins bind to ACE2 receptor, allowing for viral fusion with host respiratory epithelium. - Presentation can range from mild respiratory symptoms similar to the common cold (rhinovirus) all the way to severe respiratory disease with multi-organ failure. - Many different vaccine types exist - i.e., mRNA (Moderna; delivers mRNA coding for the spike protein), viral vector (AstraZeneca; delivers mRNA in a harmless viral capsid), and killed (Sinovac; delivers inactivated, killed virus). - Both live viral infection as well as vaccination are known to cause rare adverse effects, such as Bell's palsy and myocarditis, although these effects are not unique to coronavirus and can rarely happen with many viral infections and vaccines. - Vaccination mandates and their political implications were (and still are) a source of contentious debate. - Prior to the 2019 pandemic, coronavirus was known to cause SARS in China in 2002 and Middle Eastern Respiratory Syndrome (MERS) in Saudi Arabia in 2012. - Vaccination schedule for children now recommends IM vaccine starting at 6 months; 2-3-doses.

Vaccine stuff (2CK/3 only)	
S. pneumo	- PCV15 at 2, 4, 6 months, then again 12-15 months. - Age ≥65: PCV20 once; OR, PCV15 then PPSV23 a year later. - For **asplenia/surgical splenectomy**, sickle cell, cochlear implant, CSF leak, chronic renal failure/nephrotic syndrome, HIV, and immunosuppressed patients: extra dose of PCV15, followed by PPSV23 8+ weeks later.
Influenza	- IM killed vaccine: start age 6 months, then give yearly in the fall or winter throughout life; safe to give during pregnancy. - Intranasal live-attenuated: ages 2-45; immunocompetent, non-pregnant persons.
Coronavirus	- IM vaccine starting at 6 months; 2-3-doses.

HY Pediatric-related respiratory stuff	
Tracheoesophageal fistula (TEF)	- HY pediatric condition in which the proximal esophagus ends in a blind pouch + the distal esophagus connects to the trachea. 

	- Will present as a neonate who coughs up milk with initial feeding. - Highest yield point about TEF is that diagnosis is made via **insertion of nasogastric tube** (which cannot be inserted fully because it hits the blind pouch of the esophagus). - An NBME Q wants "endoderm" as the abnormal embryology for TEF (makes sense, since esophagus is epithelial lining of the gut → endoderm).
Choanal atresia	- Weird condition in which the nasal passages don't develop patency, so the neonate is an obligate mouth-breather.  - Will present as a child who becomes blue/hypoxic while breast feeding (because he can't breathe through the nose), then cries/becomes pink once detaching from the breast. - Similar to TEF, diagnosis is made via insertion of nasogastric tube. - Part of **CHARGE** syndrome → **C**oloboma of the eye (hole in the eye), **H**eart defects, **A**tresia of the choanae, **R**enal defects, **G**enitourinary anomalies, **E**ar anomalies.
Congenital diaphragmatic hernia	- One of the highest yield Peds conditions on USMLE. - Caused by **failure of formation of pleuroperitoneal membranes.** - Always occurs on the left, where bowel from the abdomen can herniate up into the left-chest. - Can present as ↓ bowel sounds in the abdomen + ↑ bowel sounds in the left hemithorax. 

	- Vignette might say there are cystic-appearing areas in the left hemithorax seen on CXR (loops of bowel). - Q can also say the tip of an NG tube is seen in the left hemithorax (because the stomach has herniated upward into the chest).
Neonatal respiratory distress syndrome (NRDS)	- Aka hyaline membrane disease. - The answer on USMLE for respiratory distress in kid who is born <34 weeks' gestation. - Due to insufficient surfactant production by type II pneumocytes due to ↓ lamellar bodies (the specialized organelles that produce surfactant). - These kids have ↓ lecithin/sphingomyelin ratio (i.e., <2.0). Normally it is >2-2.4. - Another name for lecithin is dipalmitoyl phosphatidylcholine. This is asked on NBME. - USMLE can give simple vignette of NRDS and then ask for various manipulation of the ratio – i.e., "↑ sphingomyelin" might be an answer (makes sense, since this would ↓ the ratio). - ↓ surfactant production means ↓ alveolar compliance and ↑ **elastic recoil.** Surfactant is hydrophobic and normally prevents the alveoli from collapsing, so if it's deficient, the hydrophobic interactions of the alveolar walls enable ↑ collapsing/elastic recoil. - CXR shows a "reticulogranular" appearance. Very buzzy and HY. - In order to prevent NRDS, a pregnant woman giving birth <34 weeks' gestation must be given ***two*** boluses of corticosteroids within 24 hours of parturition, which accelerate fetal lung maturity. For example, there is a 2CK Obgyn Q where they tell you a woman giving birth at 33 weeks' gestation was given a bolus of corticosteroids 12 hours ago. They ask for next best step → answer = "bolus of corticosteroids." Sounds weird because they said it was just done, but she needs ***two*** boluses. - Even though a neonate is preterm if born <37 weeks, we only give steroids if fetus is born 34 weeks or earlier. - For management, 2CK wants tactile stimulation first, then place under warming lights. This sequence is HY. Then give exogenous surfactant and oxygen.
Transient tachypnea of the newborn (TTN)	- The answer on USMLE when the vignette sounds like NRDS but the kid is term. For instance, they'll say neonate has difficulty breathing, but then you'll notice he's >37 weeks. - Usually seen **following C-section** or fast vaginal delivery. - Mechanism is insufficient time for the pulmonary lymphatics to clear amniotic fluid from the lungs. In addition, the pressure exerted on the fetus by the birth canal can help expulse fluid from the fetal lungs, so in the setting of a C-section, we don't have this. - CXR shows "fluid within fissure lines."
Bronchopulmonary dysplasia	- Fibrotic lung disease in an infant caused by continued use of supplemental oxygen. - Vignette will give you a kid who was born at 26 weeks' gestation who was in ICU on oxygen + now is 4 months old and is on home oxygen. They will say ECG shows right-axis deviation +/- CXR shows increased pulmonary vascularity → this reflects RVH (ensuing cor pulmonale) from pulmonary hypertension caused by the bronchopulmonary dysplasia.
Foreign body aspiration	- Answer on 2CK Peds for kid under 2 who was crawling around on the floor who now has acute-onset respiratory distress. The vignette will then add one of two features: 1) Unilateral hyperresonance in one lung (but not pneumothorax); 2) Unilateral atelectasis + ipsilateral tracheal shift.

	- Exam will ask for "bronchoscopy," or "fiberoptic examination of the airways" as the answer.
Allergic rhinitis	- Not limited to kids, but can be part of atopy in patients with asthma. - "Cobblestoning" of nasal mucosa is buzzy for allergy. This same word is used to describe the tarsal conjunctiva in allergic conjunctivitis on Peds. - NBME for 2CK has "use of pillow and mattress covers" as the Tx answer. - Otherwise, USMLE wants **intra-nasal corticosteroids** as first-line med. - USMLE does not want anti-histamines as first-line med. - 2nd-gen H1 blockers (i.e., loratadine) are the answer if *intra-nasal* corticosteroids are not listed. Oral prednisone is a **wrong** answer. - I've seen students choose prednisone and then get irascible and say, "But you said steroids were first-line." → Yeah, bro. Intra-nasal. Not fucking oral.
Adenoid hypertrophy	- Just know that enlarged tonsils (adenoids) are a known cause of stridor in pediatrics. Answer = tonsillectomy.
Peritonsillar abscess (Quinsy)	- Pus collection causing pain and swelling at posterior oropharynx. - Can be seen as complication of tonsillitis. - Patient can have **difficulty opening his/her mouth;** this contrasts with bacterial tracheitis, where the patient can fully open his/her mouth (on 2CK NBME, where this detail is important in an otherwise vague vignette). 
Laryngomalacia	- Most common cause of stridor in infants. - Softening of airway cartilage. - Vignette will give kid under the age of 1 who has noisy breathing that is mitigated when placed prone (on stomach) or upright. - Self-resolves almost always (i.e., don't treat).
Vascular ring	- Weird cause of stridor that USMLE likes to contrast with laryngomalacia. - Vascular embryology variant can lead to airway compression. - In contrast to laryngomalacia, mitigated when neck is extended. - Tx = surgery.
Velopharyngeal insufficiency	- High-pitched, hyper-nasal, or whistling speech in patients with Hx of cleft palate due to shortened palate, with or without repair. - It's the answer on a 2CK NBME, so know that it exists. They will simply just give you vignette of cleft palate + ask what the kid is at risk for.
Cholesteatoma	- Idiopathic squamous proliferation in middle ear behind the tympanic membrane that presents as an enlarging mass. - Can cause obstructive hearing loss + grow into inner ear. - Requires surgery. - Sounds like an obscure diagnosis, but it's not. Be aware it exists for Peds.
Tracheomalacia	- Not pediatric, but I'm putting it here because I see this confused with laryngomalacia.

	- Seen usually in adults who have stridor developing months post-surgery. - Caused by Hx of intubation leading to softening of airway cartilage. - Just know this diagnosis / it exists + don't confuse with laryngomalacia.

Differentiating viral vs bacterial URTI
- If you're studying for Step 1, you can basically ignore this table. If you're studying for 2CK, the CENTOR criteria are exceedingly HY for Family Med and Peds.
- If 0 or 1 point, the URTI is unlikely to be bacterial (i.e., it's likely to be viral). If 2-4 points, the chance is much greater that the URTI is bacterial. - **1) <u>Absence of</u>** cough (i.e., no cough = 1 point; if patient has cough = 0 points). - **2)** Fever >38 C. - **3)** Tonsillar exudates. - **4)** Lymphadenopathy (cervical, submandibular, etc.). - There is a version of the criteria that includes age, but on the USMLE it can cause you to get questions wrong. So just use the simplified above four points. - If 0-1 point, answer = "supportive care"; or "no treatment necessary"; or "warm saline gargle" (same as supportive care); or "acetaminophen." Latter is answer for 3M with viral URTI + fever on Peds NBME form 2. - If 2-4 points, next best step = "rapid Strep test." If rapid Strep test is negative, answer = throat culture, **NOT** sputum culture. - While waiting on the throat culture results, we send the patient home with amoxicillin or penicillin for presumptive Strep pharyngitis. - If child is, e.g., 12 years old, and develops a rash with the beta-lactam, answer = beta-lactam allergy. - If the vignette is of a 16-17 year-old who has been going on dates recently (there will be no confusion; the USMLE will make it clear) + gets a rash with the beta-lactam, the answer = EBV mononucleosis; therefore do a heterophile antibody test (Monospot test). - **EBV is the odd virus out that usually presents with all four (+) CENTOR criteria and presents like a bacterial infection.** - This is why it's frequently misdiagnosed as Strep pharyngitis. It is HY to know that beta-lactams given to patients with EBV may cause rash via a hypersensitivity response to the Abx in the setting of antibody production to the virus. EBV, in a patient who does not receive Abx, can cause a mild maculopapular rash. But the rash with beta-lactam + EBV causes a more intense pruritic response generally 7-10 days following Abx administration on the extensor surfaces + pressure points.

Ventilator settings nonsense for USMLE
- Students will sometimes nag about pedantic ventilator settings info. The vast majority of it is garbage for USMLE. What I can basically say is that **increasing PEEP** (positive-end expiratory pressure) is probably the answer 4 out of 5 times for any ventilator settings-type of question. The Q might give you a big 15-line rambling paragraph where you're not sure what's going on, and then the answer is just "increase PEEP." - PEEP keeps the alveoli open longer, thereby facilitating gas exchange. - When the patient is on a ventilator, the general aims are: - Keep FiO2 as low / as close to room air as possible (i.e., high O2 can cause free radical damage); - Keep tidal volume as low/normal as possible (i.e., 400-500-ish mL); high tidal volumes can cause ventilator-associated barotrauma. - Increase PEEP if it means lower FiO2 and tidal volumes can be achieved.

- But the above needs to be taken with a caveat. For example, the Q might say the patient is on a ventilator and has arterial pO2 of 40 mm Hg (normal is 80-100) + FiO2 is 100% + tidal volume is 1000 mL, and the answer will be "increase PEEP." The student says, "Wait but I thought you said we want to reduce FiO2 and tidal volume." → Sure, but we can't do that here because pO2 is super-low, so that would only exacerbate it further.
- As I talked about for ARDS earlier, you could be aware the triad of 1) prone positioning, 2) low-tidal volume mechanical ventilation, and 3) permissive hypercapnia can be implemented.
- A very HY point is that ↓ CO2 causes ↓ cerebral perfusion (this is why patients faint in panic attacks, which is asked on NBME). The first step in managing ↑ intracranial pressure is "intubation + hyperventilation."
- They also ask the inverse of this on NBME – i.e., they say an anesthesiologist wants to ↑ a patient's cerebral perfusion, and the answer is "decrease respiratory rate" → causes ↑ CO2 → ↑ cerebral perfusion.
- Ventilator-acquired pneumonia (VAP) should be treated with vancomycin PLUS either ceftazidime or cefepime. Vancomycin not only covers MRSA but also is effective against some high-resistance *S. pneumo* strains. Ceftazidime (3rd generation ceph) and cefepime (4th gen ceph) are very effective against *Pseudomonas*, which is a HY nosocomial organism for VAP.
- Patients should ideally be weaned from ventilators as quickly as possible. USMLE also likes patients who are sick on ventilators as sometimes having euthyroid sick syndrome – i.e., they will say patient cannot be weaned from ventilator + has ↓ T3 and normal TSH → answer = euthyroid sick syndrome. The full array of arrows are: ↓ T3, ↑ rT3, ↔ T4, ↔ TSH. If you have no idea what I'm talking about, then I recommend going through my HY Arrows PDF.
- If a patient ever has a high CO2 (NR 33-44 mmHg) → answer can = "ventilatory insufficiency." This applies to both patients on and off ventilators. Don't be confused if they give you ↑ respiratory rate. New Surg form gives RR of 40 with ↑ CO2 as an example. Sometimes patients have shallow breathing despite ↑ RR.

HY Pulmonary drug points	
Bosentan	- Endothelin-1 receptor antagonist used for pulmonary hypertension. - Endothelin is a vasoconstrictor ↑ in pulmonary hypertension. - There is an NBME Q where guy is shaving + cuts himself + they ask what will be seen at site of injury → answer = ↑ endothelin. So be aware that even though 14/15 times this relates to pulmonary vascular constriction, it is technically a vasoconstrictive mediator not isolated to the lungs.
Methotrexate	- Dihydrofolate reductase inhibitor. - 1st-line DMARD for RA + 1st-line oral agent for psoriasis that fails topicals. - Causes pulmonary fibrosis (and hepatotoxicity and neutropenia). - Toxicity can be mitigated with "leucovorin rescue," aka folinic acid (not folic acid).
Albuterol	- Short-acting β2-agonist used for asthma. - Can cause tremor (asked on a Step 1 NBME).
Fluticasone, Beclomethasone	- Inhaled corticosteroids (ICS) used for asthma and COPD. - Used for prevention / to decrease recurrence of episodes. They have no role in acute exacerbations. For the latter, IV methylprednisolone is used.
Prednisone	- Most effective at ↓ recurrent asthma attacks, but used last resort because oral steroids can cause Cushing syndrome and growth stunting.
Montelukast, Zafirlukast	- Leukotriene receptor antagonists (LTC, D, and E4). - Leukotrienes normally cause bronchoconstriction. So these agents ↓ airway constriction. - LTB4 is special and is a neutrophilic chemotactic molecule. For immuno, you can memorize that LTB4, IL-8, and C5a are HY neutrophilic chemotactic molecules. - Can be used for asthma in patients with aspirin-induced asthma (blockage of COX shunts arachidonic acid down the lipoxygenase pathway toward leukotrienes), or for asthma in general in between a LABA and prednisone.
Zileuton	- Lipoxygenase inhibitor. Blocks synthesis of leukotrienes. - Use-case same as the -lukasts.
Nedocromil, Cromolyn sodium	- Mast cell stabilizers used for asthma. Prevent release of histamine.

	- Step 1 form has a Q where they ask, "Which of the following agents blocks the release of an autocoid from its cellular storage site?" → answer = cromolyn sodium. - Autocoids are very brief-acting hormones that "are made, play their role, and are destroyed at the same site." → Histamine, serotonin, bradykinin, and angiotensin are examples.
Tiotropium	- Long-acting muscarinic receptor antagonist (LAMA) used first-line for COPD.
Olodaterol	- Long-acting β2-agonist (LABA) used first-line for COPD. - As discussed earlier, either a LAMA or LABA can be used first-line for COPD.
Ipratropium	- Short-acting muscarinic receptor antagonist (SAMA) used for COPD. - Not first-line anymore, but still shows up on Family Med material as an agent that's used. Some old-school practitioners still use it, so don't disregard it.
Epoprostenol	- PGI2 prostacyclin that can be used for pulmonary hypertension.
Sildenafil	- Viagra; PDE-5 inhibitor that can also be used for pulmonary hypertension.
Nifedipine	- Dihydropyridine calcium channel blocker that can be used for pulmonary HTN. - As discussed in the Cardio PDF, can cause peripheral edema/fluid retention.
Omalizumab	- Monoclonal antibody against IgE. - Can be used in theory in severe asthmatics who have high IgE levels.
Dornase alfa	- Nucleotidase used to break down airway mucous in cystic fibrosis patients.
Guaifenesin	- Mucolytic agent that softens mucous in cystic fibrosis.
Ivacaftor	- Helps localize CFTR channel to membrane and correct its folding (in CF clearly).
Pirfenidone	- Used for idiopathic pulmonary fibrosis. - Anti-fibrotic agent that inhibits TGF-β-mediated collagen synthesis.

- 58F + 6-month Hx of shortness of breath and non-productive cough + 2-year Hx of worsening dysphagia + 20-yr Hx of hands turning white when exposed to cold; what lung condition is this patient most at risk of developing? → answer = **pulmonary hypertension** → pulmonary fibrosis common in CREST syndrome (scleroderma; limited systemic sclerosis) → pulmonary fibrosis seen in both diffuse and limited types of systemic sclerosis; the latter is sans the renal phenomena) → **can lead to cor pulmonale,** which is right-heart failure secondary to a pathology of lung etiology (i.e., the left heart must not be causative).
- What will the USMLE frequently say in the Q if they're hinting at pulmonary hypertension? → HY vignette descriptors are loud S2 or P2 (pulmonic valve slams shut when the distal pressure is high); dilation of proximal pulmonary arteries; increased pulmonary vascular markings (congestion + increased hydrostatic pressure).
- When is "pulmonary hypertension" important as the answer? → notably in Eisenmenger syndrome and cor pulmonale.
    - Regarding Eisenmenger: the reversal of the L-to-R shunt across the VSD such that it's R-to-L requires the tunica media of the pulmonary arterioles to hypertrophy secondary to chronic increased preload → proximal backup of hydrostatic pressure → increased afterload on the

right heart → *now* the right heart starts to significantly hypertrophy → shunt across the VSD reverses R-to-L. This is important, as RVH is not the most upstream cause of Eisenmenger; the pulmonary hypertension is. Regarding cor pulmonale: as mentioned above, cor pulmonale is right heart failure secondary to a pathology of pulmonary etiology (e.g., COPD, cystic fibrosis [CF], fibrosis, etc.). The most common cause of right heart failure is left heart failure, so for cor pulmonale to be the diagnosis, the left heart **must not** be the etiology of the right heart failure; the lungs must be the etiology. **Pulmonary capillary wedge pressure (PCWP) must be normal.** If the cause of the right heart failure is, e.g., chronic bronchitis or CF, then it is **the degree of hypoxic vasoconstriction** that determines the prognosis for the cor pulmonale. For chronic bronchitis and CF --> hypoxic vasoconstriction of pulmonary vessels --> increased afterload in the proximal pulmonary arteries --> dilatation and congestion of proximal pulmonary arteries --> more hydrostatic pressure in the proximal pulmonary arteries --> increased afterload experienced by the right heart --> right heart hypertrophies --> right heart eventually decompensates secondary to chronically elevated pulmonary vascular pressure --> classic right heart failure signs ensue (JVD + pulmonary edema; hepatosplenomegaly may also be seen [e.g., spleen tip and/or liver edge are palpable beneath the costal margin]).For emphysema, loss of alveolar surface area --> loss of capillary surface area --> increase in pulmonary vascular resistance (decreased # of capillaries forming parallel circuit) --> pulmonary hypertension --> increased afterload on right ventricle --> cor pulmonale.

- "Please explain PCWP. I've seen that before but no idea what that means or how it applies to questions." → If you stick a catheter into the venous circulation and feed it all the way back up to the right heart and into the pulmonary circulation, eventually it won't be able to go and farther and will be lodged in a distal pulmonary arteriole; the pressure reverberations sensed by the catheter must therefore reflect the pulmonary capillary pressure (immediately distal to the catheter); but of course the pulmonary capillary pressure must reflect the left atrial pressure because the latter is immediately contiguous; **therefore PCWP = left atrial pressure.** This is a very simple rule that must be memorized and understood for the USMLE. For the purpose of questions, not only will PCWP be normal in the setting of cor pulmonale, but it will also be elevated if there is any left heart pathology.

  - Hypovolemic shock arrows → CO down, VR down, TPR up, **PCWP down** (or low-normal).
  - Cardiogenic shock arrows → CO down, VR down, TPR up, **PCWP up.**
  - Septic + anaphylactic shock arrows → CO up, VR up, TPR down, **PCWP normal.**
  - Neurogenic shock + adrenal crisis arrows → CO down, VR down, TPR down, **PCWP normal.**

- 23F + non-smoker + gradually worsening shortness of breath + loud P2 + CXR shows increased pulmonary vascular markings; Dx? → answer = primary pulmonary hypertension → common cause is mutations in BMPR2.
- Treatment of pulmonary hypertension? → USMLE will generally not make you pick and choose between agents, however it is generally accepted in the literature that **dihydropyridine calcium channel blockers,** such as nifedipine, are effective in vasoreactive pulmonary hypertension. Other agents include prostaglandin I2 (epoprostenol), **bosentan (endothelin-1 receptor antagonist),** and phosphodiesterase-5 inhibitors (sildenafil; yes, Viagra).
- "Am I supposed to know something about endothelin-1?" → Step 1 pulmonary material has an obsession with it. Be aware that endothelin-1 is a potent vasoconstrictor → so increased endothelin-1 activity is associated with pulmonary hypertension.
- 23F with primary pulmonary hypertension; Q asks which of the following might be seen in this patient; answer = "increased endothelin-1 activity" → essentially, if bosentan is a HY antagonist of endothelin-1 as a Tx for pulmonary hypertension, then it makes sense that increased activity is correlated with pulmonary hypertension in patients.
- 68M + S3 + crackles in both lungs + JVD + peripheral edema; what would be seen in this patient? → answer = "increased endothelin-1 activity" → congestive heart failure is defined as right heart failure caused by left heart failure; because the lungs need to accommodate more fluid secondary to the backup from the decompensated left heart, the pulmonary vessels constrict (endothelin-1 activity) → pulmonary hypertension → right heart experiences increased afterload and then also fails.
- Which lung cancers are apical vs central? → Central → sounds like **S**entral → **S**quamous cell, **S**mall cell; peripheral → adenocarcinoma, large cell.
- 45F + non-smoker + apical lung mass; no other information; most likely cancer? → answer = adenocarcinoma → classically female non-smokers with peripheral lesion, although I've seen it show up in a male as well on NBME.

- 45F + non-smoker + peripheral lung mass + proximal muscle weakness + increased serum CK + rash on shoulders; Dx? → paraneoplastic dermatomyositis secondary to bronchogenic adenocarcinoma.
- 45F + non-smoker + lung mass + thrombosis; lung cancer Dx? → answer = adenocarcinoma. In general, cancer is associated with hypercoagulable state, but this is particularly the case for adenocarcinomas; in addition, Trousseau sign of malignancy (migratory thrombophlebitis) is characteristic of adenocarcinomas (**not** limited to head of pancreas adenocarcinoma).
- 45F + long smoking Hx + shooting groin pain; Dx? → ureterolithiasis secondary to hypercalcemia from bronchogenic squamous cell carcinoma PTHrp (PTH related peptide) secretion.
- 45F + long smoking Hx + shooting groin pain + serum studies show high calcium; what is the patient's PTH level? → answer = decreased (suppressed in the setting of high PTHrp).
- 45F + dies of lung cancer + Q shows you gross pathologic specimen of large, cavitating, central lesion; Dx? → answer = squamous cell carcinoma → cavitations are common.
- 45F + lung cancer + Hx of stabbing flank pain a couple months ago + today presents with confusion; Dx? → hypercalcemic crisis → squamous cell carcinoma with PTHrp secretion with Hx of nephrolithiasis; high calcium can cause delirium.
- 45F + long smoking Hx + central coin lesion on CXR + violaceous abdominal striae + potassium level of 3.0 mEq/L; Dx? → answer = small cell carcinoma with ectopic ACTH secretion; student says, "Wait but why's the potassium low?" → effect of chronically high cortisol (can act to secrete K at distal kidney similar to aldosterone; only seen in ongoing Cushing syndrome).
- 45F + confirmed Dx of early-stage small cell lung cancer + wobbly gait; Q asks the mechanism for her presentation; answer = autoimmune → paraneoplastic anti-Hu and anti-Yo antibodies causing cerebellar dysfunction; Dx is small cell cerebellar dysfunction (small cell is known to cause neuronal paraneoplastic syndromes beyond Lambert-Eaton).
- 45F + central lung lesion + confusion + increased urinary osmolality; Dx? → SIADH secondary to small cell → low serum sodium + high urinary osmolality; should be noted that sodium derangement (high or low), as well as high calcium, can cause CNS disturbance / delirium.
- 45F + long smoking Hx + difficulty getting up from chair but is successful after multiple attempts; what is the location of her pathology? → answer = neuromuscular junction → small cell paraneoplastic Lambert-Eaton syndrome → antibodies against presynaptic voltage-gated calcium channels.

- 45F non-smoker + episodic flushing, wheezing, and dyspnea + coin lesion seen on CXR; Dx? → answer = bronchial carcinoid tumor.
- 33F + Hx of recent molar pregnancy + very high serum beta-hCG levels + new-onset non-productive cough; CXR shows multiple peripheral densities in right lobe; Dx? → metastatic choriocarcinoma (loves to metastasize to the lungs; chorio can progress from hydatidiform mole).
- Do we do lung cancer screening? → Annual low-dose chest CT in patients who meet all of the following:
    - Age 50-80.
    - >20-pack-year Hx of smoking.
    - Have smoked in past 15 years.
- 3M + swallows a peanut; where in the lung does it go? → superior segment of right lower lobe if patient is supine; goes to inferior segment of right lower lobe if patient is upright. This is because the right mainstem bronchus is shorter and straighter than the left.
- "How does this type I vs II respiratory failure stuff apply more specifically to questions?" → Type I is when O2 and CO2 are both low. Type II is when O2 is low and CO2 is high.
    - For *acute* asthma, the patient will have a type I respiratory failure: **low pO2, low pCO2, normal bicarb, high pH** → in this setting, RR is high (e.g., 28/min), so pCO2 is low, not high, because the CO2 can diffuse out with no problem; however O2 is low because the bronchospasm and secretions prevent the slow-diffusing O2 from sufficiently entering the capillary beds; bicarb is unchanged **because it's the acute setting;** it will not normally start to appreciably decrease for at least 12-24 hours (just think, similarly, it takes at least a day for bicarb to decrease when one goes to high altitude, resulting in altitude sickness); pH is high because of acute respiratory alkalosis (once again, bicarb hasn't decreased to compensate).
    - Classic USMLE scenario for type II is asthma attack after 30-45 minutes, where the patient's values are as follows: **low pO2, normal pCO2, normal bicarb, normal pH;** the Q asks what you should do: answer = intubation → reason being: the low O2 clearly means the patient is still in respiratory distress, however the rising of CO2 into the normal range (and restoration of pH to normal) indicates this is only because *the patient is getting tired* – i.e., RR is falling out of fatigue, not because the patient's O2 has stabilized. In this setting, the patient is *in*

transition to a type II respiratory failure. In other words, if we don't intubate, the patient's values will soon become: **low pO2, high pCO2, normal bicarb, low pH** (i.e., respiratory acidosis; type II respiratory failure).

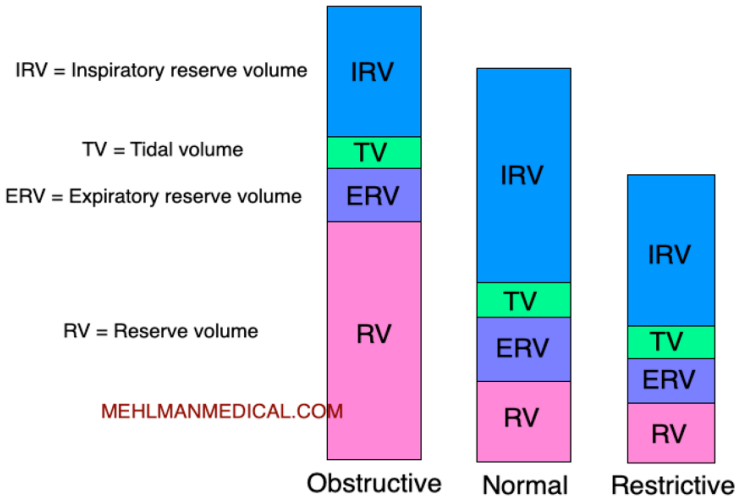
  - For pulmonary embolism, we also have a type I respiratory failure: **low pO2, low pCO2, normal bicarb, high pH,** same as acute asthma. Bicarb is once again unchanged because it's the acute setting; pCO2 is low because RR is high; O2 is low because of pathologic dead space (high V/Q → despite good ventilation, reduced perfusion results in reduced arterial oxygenation).
  - For opioids, we expect hypoventilation, so we have: **low pO2, high pCO2, normal bicarb, low pH** (respiratory acidosis; type II respiratory failure), secondary to the patient's low RR.

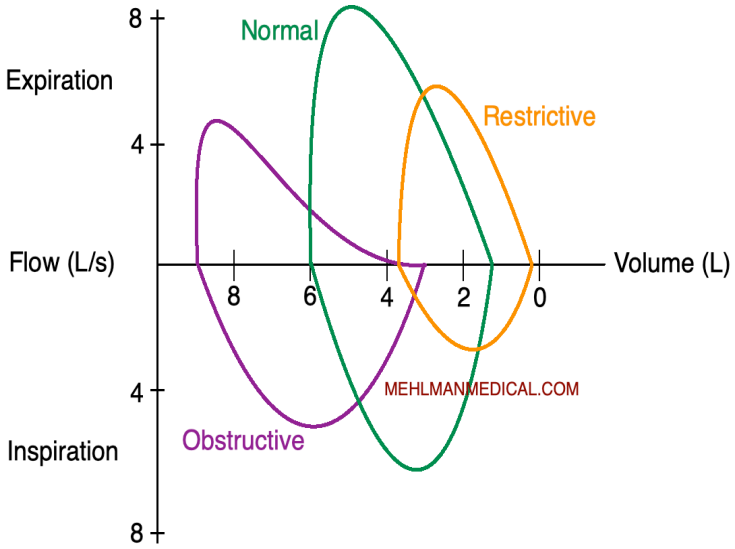
- What kind of acid-base disturbance do we have in COPD exacerbation?→ Acute on chronic respiratory acidosis. NBME can write this as "chronic respiratory acidosis with acute respiratory acidosis." Patients with COPD are chronic CO2 retainers so will have chronic respiratory acidosis (high CO2, high bicarb to compensate, low or normal pH). Then when there is an exacerbation on top of it, CO2 will shoot even higher, where have super-high CO2, high bicarb, low pH. This can occur even if respiratory rate is high.
- What do I need to know about FEV1/FVC? → normal ratio is considered >0.7 (according to AAFP); ratio is decreased in obstructive lung disease (<0.7; <70%); ratio is *normal or* increased in restrictive lung disease (>0.7; >70%). In both obstructive and restrictive, FEV1 and FVC, *as independent variables,* both decrease; it's only the ratio that differs.
- Why is the FEV1/FVC ratio normal or increased in restrictive? → typical student response is, "Well FEV1 doesn't go down as much as FVC in restrictive." But *why* is this the case? → answer that pops up all over NBME exams is **radial traction** → i.e., if the restrictive pathology is caused by fibrosis / scar formation, we could colloquially say that fibrosis is "sticky," where it sticks to the outside of the airways and keeps them from closing → the amount of air one breathes out in one second (FEV1) doesn't go down as much as it does in an obstructive lung disease, where radial traction doesn't play a role (so FEV1 goes down even more in obstructive pathologies → FEV1/FVC is low in obstructive).
- 59F + advanced COPD and lung cancer + requires partial pneumonectomy + FEV1 of left lung is 600mL; next best step? → answer = "measurement of FEV1 of the right lung" → seemingly weird

answer but HY for pulm component of surg shelves → minimum acceptable postoperative FEV1 for both lungs is 800mL.

- "What is obstructive vs restrictive lung disease?" → obstructive lung disease means air-trapping occurs because there is literally obstruction in the airways, resulting in prolonged expiratory phase; restrictive means there is decreased ability for the lungs to expand (decreased compliance) but expiratory component remains unaffected in pure restrictive pathology.
- What are some common obstructive vs restrictive conditions?
    - Obstructive: emphysema + bronchitis (combo = COPD); asthma.
    - Restrictive: autoimmune diseases (many conditions can result in fibrosis, such as rheumatoid arthritis [rheumatoid lung], systemic sclerosis [both diffuse and limited {CREST}], sarcoidosis); Hx of radiation to the chest (i.e., Hx of Hodgkin) leading to fibrosis; pneumoconioses (asbestosis, silicosis, etc.); **ankylosing spondylitis; obesity; neuromuscular disease** (i.e., myasthenia gravis, Guillain-Barre).
- What is definition of chronic bronchitis? → productive cough every day for >3 months in a year for at least two consecutive years; most common cause is smoking.
- 57M + 4-day Hx of productive cough + COPD managed with ipratropium inhaler + afebrile + HR 110 + RR 22 + BP 130/80 + O2 sats 90% + breath sounds decreased bilaterally + CXR shows hyperinflation of both lungs; Dx? → answer on NBME = "bronchitis" → i.e., acute bronchitis; usually viral.
- Classic way to differentiate COPD from asthma? → in asthma, spirometry showing decreased respiratory function is *largely reversible* with albuterol; this is modestly effective at best in COPD; "flattened diaphragm" is classic CXR descriptor for hyperinflated lungs seen in COPD; this descriptor can be seen in asthma, but is rare.
- "What do I need to know about DLCO?" → stands for diffusion capacity of the lung for carbon monoxide (correct, not dioxide); test measures difference in partial pressures of inspired vs expired carbon monoxide; you just need to know that DLCO is decreased for most lung pathologies, **however it is increased in asthma;** how this applies to questions → if you get a hard vignette (e.g., COPD vs asthma, etc.) and they tell you DLCO is down, you can say, "Ok cool, it's not asthma."

- "What do we need to know in terms of lung volumes and flow-loop stuff for obstructive vs restrictive? → in general, lung volumes are increased in obstructive lung pathologies and decreased in restrictive ones. The following diagrams illustrate some of the changes in obstructive vs restrictive lung disease:





- 72F + longitudinal study comparing her lung function to 20-yr-olds' lungs; what changes would we expect in her versus young individuals? → answer = **residual volume ↑; arterial pO2 ↓; A-a pO2 gradient ↑;** essentially, people develop obstructive lung disease changes as they age (likely due to lifelong inhalation of particulates, akin to smoking).
- 38F smoker + study performed comparing her to healthy individuals; what changes would we expect in her compared to non-smokers → answer = **mucous production and secretion ↑; activity of airway cilia ↓; alveolar macrophage function ↓;** increased mucous production and secretion reflects changes tending toward bronchitis; activity of airway cilia *decreases* with exposure to particulates; this point is

insidiously HY for USMLE Step 1; alveolar macrophage function and activity are impaired by particulates (even though increased clearance necessary).

- 59M + 70-pack-yr Hx of smoking + Q asks which cell is likely to be abnormal in this patient; answer = pseudostratified columnar epithelial cells; wrong answers are alveolar macrophages, type I/II pneumocytes, alveolar endothelial cells; once again, USMLE wants you to know cilia function is decreased; pseudostratified columnar epithelial cells of the respiratory tract are ciliated.
- What is one of the most important descriptors in a Q for restrictive lung disease? → **reticulonodular or reticular** pattern seen on CXR and CT → memorize these descriptors as = restrictive lung disease; they are also known as **honeycombing** colloquially.
- "When is centri-acinar vs pan-acinar emphysema important?" → you just need to know that centri-acinar is the distribution seen in smoking; pan-acinar is seen in alpha-1 anti-trypsin deficiency.
- 32M non-smoker + reduced lung function + has a 40-year-old brother with COPD + father died of alcoholic liver disease; Dx? → answer = alpha-1 anti-trypsin deficiency → enzyme is produced in liver and travels to the lungs; functions to breakdown elastase; deficiency results in pan-acinar emphysema (Q will mention non-smoker with emphysema); can also result in hepatic cirrhosis (alcohol exacerbates already poor liver prognosis); codominant inheritance.
- "When is bronchiectasis the answer?" → when the vignette tells you "cups and cups of foul-smelling sputum"; this is how bronchiectasis presents 8 out of 10 times; there is destruction of the musculature of the bronchioles resulting in ectatic (dilated) airways → "loss of elastic support to the walls of the bronchioles" (NBME); Q might also show you a CT scan (1 out of 10 Qs) where it appears as though there's dilated spaces throughout both lung fields; most common cause worldwide is TB; in western countries it's CF; smoking is also ubiquitous cause; the hardest presentation is pediatrics (1 out of 10 Qs), where the Dx is "right middle lobe syndrome" → Q will say there's a 6-12-month Hx of non-productive cough + a linear/streaky opacity seen on CXR in the right middle lobe → answer = bronchiectasis. Student will say "wtf? I thought it was supposed to be cups and cups of sputum." Yeah, it normally is, but now you know of another presentation.
- 34F + non-productive cough + CXR shows hilar nodularity; Dx? → answer = "non-caseating granulomas" → sarcoidosis; Tx = steroids; CXR will show bihilar lymphadenopathy; non-caseating

granulomas produce 1-alpha hydroxylase, which converts inactive 25-OH-D3 into active 1,25-(OH)2-D3, leading to increased absorption of Ca and PO4 in the small bowel; Q will sometimes give PO4 in the normal range; also need to know that *fecal calcium* is low (asked on NBME; makes sense if you absorb it); PTH is also suppressed.

- Any drug that can be used to help diagnose asthma? → answer = methacholine (muscarinic agonist); will reproduce asthma Sx; never use during acute attacks.
- 34F + non-productive cough + CXR normal; Dx? → answer = "activation of mast cell" → asthma.
- Asthma Tx (outpatient)? → albuterol (short-acting beta-2 agonist; SABA) inhaler for immediate Mx → if insufficient, start low-dose ICS (inhaled corticosteroid) preventer → if insufficient, maximize dose of ICS preventer → if insufficient, add salmeterol inhaler (long-acting beta-2 agonist; LABA); in other words:
    - **1) SABA; then**
    - **2) low-dose ICS; then**
    - **3) maximize dose ICS; then**
    - **4) LABA.**
    - The above order is universal. After the LABA, any number of agents can be given in any order – i.e., nedocromil, cromolyn sodium, zileuton, montelukast, zafirlukast, etc.
    - Last resort is **oral corticosteroids, however they are most effective.** In other words:
- 12M + ongoing wheezing episodes + is on albuterol inhaler; next best step? → answer = add low-dose ICS.
- 12M + ongoing wheezing episodes + is on albuterol inhaler; what's most likely to decrease recurrence? → answer = oral corticosteroids (student says "wtf? I thought you said ICS was what we do next and that oral steroids are last resort" Yeah, you're right, but they're still **most effective** at decreasing recurrence. This isn't something I'm romanticizing; this distinction is assessed on the FM NBME forms.
- MOA of nedocromil and cromolyn sodium? → answer = mast cell stabilizers.
- MOA of zileuton? → answer = lipoxygenase inhibitor (enzyme that makes leukotrienes from arachidonic acid).

- MOA of the -lukasts? → answer = leukotriene LTC, D, and E4 inhibitors. LTB4 receptor agonism is unrelated and induces neutrophilic chemotaxis (LTB4, IL-8, kallikrein, platelet-activating factor, C5a, bacterial proteins).
- 16M goes snowboarding all day + takes pain reliever for sore muscles afterward + next day develops wheezing out on the slopes again; what's going on? → answer = took **aspirin + this is Samter triad** (now cumbersomely known as aspirin-exacerbated respiratory disease [AERD]) → triad of aspirin-induced asthma + aspirin hypersensitivity + **nasal polyps**). Just to be clear, other NSAIDs can precipitate Samter triad, but the literature + USMLE will make it explicitly about aspirin.
- 16M takes aspirin + gets wheezing; what are we likely to see on physical exam? → answer on USMLE = nasal polyps.
- "Wait I don't understand. Why would aspirin cause asthma?" → arachidonic acid can be shunted down either the cyclooxygenase or lipoxygenase pathways; if you knock out COX irreversibly by giving aspirin (or reversibly with another NSAID), more arachidonic acid will be shunted down the lipoxygenase pathway → more leukotrienes → more bronchoconstriction.
- Kid has Hx of AERD; physician considers agent to decrease his recurrence of Sx → zileuton, or -lukasts (both are correct; and only one will be listed).
- 10F + diffuse wheezes heard bilaterally + sputum sample shows numerous eosinophils and Charcot-Leyden crystals; which of the following is implicated in her disease pathogenesis? → answer = leukotriene C4 (LTC4); should be noted that LTC4, LTD4, and LTE4 all cause bronchoconstriction; LTB4 stimulates neutrophilic chemotaxis; Charcot-Leyden crystals are seen in asthma and are composed to eosinophil protein galectin-10.
- Any weird asthma Txs? → **omalizumab** → monoclonal antibody against IgE → used for intractable, severe asthma unresponsive to oral steroids + in patients who have eosinophilia + high IgE levels (I asked a pulmonologist about this drug years ago when I was in MS3 and he said he was managing 1000 patients with asthma and just three were on omalizumab).
- Acute asthma Mx (emergencies) → most important piece of info straight-up is: USMLE wants you to know that inhaled corticosteroids (ICS) have **no role in acute asthma management.** First thing we do is give oxygen (any USMLE Q that shows depressed O2 sats, answer is always O2) + nebulized

albuterol (face mask with mist); **IV steroids** are then administered. The Mx algorithm is more complicated, but that is what you need for the USMLE.

- 25F + runny nose and eyes in spring + cobblestoning of nasal mucosa; Dx + Tx? → answer = allergic rhinitis ("cobblestoning" is buzz term but used on the NBME); Tx = avoidance of precipitating allergen (if known, e.g., pollen); if meds used, first-line = intranasal corticosteroids; second-line is oral or intranasal antihistamines (e.g., loratadine).
- 27M + inhalation injury from housefire + Q asks "destruction to which structure is most likely to preclude restoration of normal pulmonary architecture and lung function?" → answer = basement membranes → USMLE wants you to know intact basement membranes necessary for complete healing from lung injury or infection.
- 44M alcoholic + fever + air-fluid level seen on CXR; Dx + TX? → answer = pulmonary abscess; Tx = clindamycin; air-fluid level = circle on CXR, where bottom half is pus (radiopaque; white); top half is air (radiolucent; black); due to aspiration of "oropharyngeal anaerobes" (bacteroides, Mobiluncus, Peptostreptococcus).
- 44M alcoholic + fever + CXR shows lobar consolidation + sputum is thick and red; Dx? → *Klebsiella pneumoniae* (produces thick, mucoid colonies with "currant jelly sputum"); common cause of aspiration pneumonia (increased risk in alcoholics).
- 44M alcoholic + fever + CXR shows lobar consolidation + extremely foul-smelling sputum; Dx? → answer = *Bacteroides* pneumonia.
- 44M alcoholic + fever + air-fluid level seen on CXR + clindamycin is given and resolves his condition + months later he has residual lesion seen on CXR; why? → answer = failure of maintenance of basement membranes.
- Function of type I vs II pneumocytes? → type I composes 95% of alveolar surface area (simple squamous) and is responsible for gas exchange; type II pneumocytes are interspersed and fewer; they produce surfactant (contain specialized surfactant-producing organelles called lamellar bodies) and also function as the stem cells of the lung (i.e., after lung injury, type II multiply and restore type I cells, insofar as basement membranes remain intact).
- What are Clara cells → secrete glycosaminoglycans to protect the lining of bronchioles.

- Child born at 28 weeks gestation + dyspnea + Q asks which of the following is likely true in the patient → answer = "deficiency of lamellar bodies" → neonatal respiratory distress syndrome (NRDS); aka hyaline membrane disease → deficiency of surfactant → reduced lecithin (dipalmitoyl phosphatidylcholine) to sphingomyelin ratio; ratio should be >2.
- Child born at 28 weeks gestation + dyspnea + Q asks which of the following is likely true in the patient → answer = deficiency of surfactant protein D → need to know this is decreased in NRDS.
- Neonate born at 38 weeks gestation via C-section + RR of 70 (normal 40-60) + CXR shows bilateral mild hyperinflation and prominent perihilar interstitial markings ; Dx? → answer = transient tachypnea of the newborn → TTN is the answer when the vignette "sounds like NRDS but the kid is term"; **seen in C-sections and fast vaginal deliveries in term neonates;** mechanism is delayed absorption of the fetal lung fluid by pulmonary lymphatics; Tx is supportive.
- Neonate born at 26 weeks + required oxygen in ICU for several weeks + is now on home oxygen; child is at increased risk for what? → answers = bronchopulmonary dysplasia, retinopathy of prematurity, germinal matrix bleed → all can be seen in neonates on oxygen therapy.
- Newborn + decreased bowel sounds in abdomen + tracheal shift to the right; Dx? → answer = congenital diaphragmatic hernia → incomplete formation of pleuroperitoneal membranes → occurs on the left side, with bowel herniating into the left hemithorax.
- Newborn + bowel sounds heard in the left chest; mechanism? → answer = incomplete formation of pleuroperitoneal membranes.
- 12-hour-old newborn + excessive oral secretions and coughing after first feed + Sx resolved after suctioning + pregnancy was characterized by polyhydramnios; next best step in Mx? → answer = insertion of a nasogastric tube → Dx = tracheoesophageal fistula.
- 12-hour-old newborn + becomes blue when breastfeeding + becomes pink again when crying; Dx? → atresia of the choanae (CHARGE syndrome → Coloboma of the eye, Heart defects, Atresia of the choanae, Retardation, Genitourinary abnormalities, Ear abnormalities/deafness).
- Can you comment on HY pneumoconioses? → lung disease secondary to occupational exposures.
    - Asbestosis → workers in shipbuilding, plumbing, construction/insulation, roofing industries → causes **supradiaphragmatic and pleural plaques** (described as "soft tissue density" visualized on CXR on 2CK NBME); plaques can calcify; biopsy may show **ferruginous bodies**

(due to deposition of hemosiderin by alveolar macrophages attempting to digest asbestos fibers); Step 1 NBME Q asks which cell is responsible for the pulmonary fibrosis seen in asbestosis → answer = macrophage.

- Anthracosis → "coalminer's lung"; carbon deposition; black appearance grossly; also what city-dwellers may develop by old age; restrictive lung pattern on spirometry.
- Silicosis → stone quarry workers; eggshell calcifications; increased risk of TB due to decreased alveolar macrophage function; do not give anti-TNF-alpha agents to these patients (because they further increase risk of TB); do PPD test if new Dx of silicosis; restrictive on spirometry.
- Berylliosis → aeronautical industry workers; granulomatous disease, so steroids might provide benefit; steroids do not help in the other pneumoconioses.
- Byssinosis → pneumoconiosis caused by hemp / textile fibers.
- Bagassosis → pneumoconiosis caused by sugarcane (specific antigen = thermophilic Actinomycetes).
- Caplan syndrome → pneumoconiosis + rheumatoid arthritis (restrictive).

- 34M hay farmer + progressive dyspnea over many months + clubbing; Dx? → answer = hypersensitivity pneumonitis; can be acute, subacute, or chronic; occurs in response to various antigens, including thermophilic Actinomycetes (hay farmers; sugarcane); non-tuberculous mycobacterium **(hot tub lung; on NBME);** bird fancier's lung, coffee worker's lung, etc.; characterized by multinucleated giant cells on biopsy.
- 28M + asthma + recurrent lung infections + skin allergy testing shows hypersensitivity to aspergillus antigen; Dx? → allergic bronchopulmonary aspergillosis (ABPA); usually seen in patients who have asthma or cystic fibrosis.
- 34M + HIV positive + nodular density seen in right upper lobe on CXR; next best step? → answer = "biopsy of the mass" → Aspergilloma (fungus ball); increased risk with Hx of TB (can occupy cavities).
- 57M + Hx of dermatomyositis + few weeks ago had fever and pneumonia + presentation hasn't resolved with multiple antibiotics + multiple sputum cultures are negative; Dx? → answer = cryptogenic organizing pneumonia (COP; formerly known as bronchiolitis obliterans organizing pneumonia [BOOP]); → frequently presents as pneumonia-like presentation (i.e., fever, dyspnea) in

patients with Hx of autoimmune disease like dermatomyositis or rheumatoid arthritis; CXR resembles atypical pneumonia; COP diagnosis suspected after failure of resolution with multiple Abx and negative sputum cultures; CT shows reverse-halo sign in 20%; biopsy showing Masson bodies confirms Dx; Tx = steroids. "Organizing" refers to persistence of alveolar exudates from a pneumonia that ultimately fibrose.

- 43F + lung transplant several months ago + declining lung function + CXR shows mild flattening of the diaphragm + CT shows air-trapping; Dx? → bronchiolitis obliterans (constrictive bronchiolitis; "popcorn lung"); do not confuse with COP (BOOP); bronchiolitis obliterans is the answer for obstructive lung disease that progressively manifests post-lung transplant (75%) or bone marrow transplant; also associated with toxic fumes and E-cigarettes; biopsy confirms Dx; irreversible, but steroids may help.
- 43F + rheumatoid arthritis + progressive shortness of breath over six months + CT of chest shows reticulonodular pattern; Dx? → answer = "usual interstitial pneumonia" (UIP); student says wtf? → on the NBME; this Dx refers to a patient with scarring and fibrosis of the lungs; if the cause is idiopathic, we call the UIP "idiopathic pulmonary fibrosis," but IPF is still UIP. Bottom line is: be aware of the term "usual interstitial pneumonia" as synonymous with pulmonary fibrosis.
- Drugs causing pulmonary fibrosis? → **methotrexate** (patients with RA who have UIP → hard to know whether it's from the methotrexate or rheumatoid lung, or both); amiodarone, bleomycin, busulfan, nitrofurantoin.
- 10-month-old + fever of 101F + wheezes bilaterally; DX + Tx? → RSV bronchiolitis → Tx is supportive; answers such as ribavirin and palivizumab are almost always wrong on actual NBME assessment.
- 8M + leaning forward in tripod position + drooling; Dx + Tx? → answer = epiglottitis caused by *Haemophilus influenzae type B*; Tx = immediate intubation; neck x-ray shows thumbprint sign; even if kid is stable, still intubate (airway can inflame + obstruct at any moment); give ceftriaxone to patient; give rifampin to close contacts; usually seen in immigrants due to lack of adequate vaccination.
- 8M + hoarse cough + improves when father brings him out in the cold; Dx + Tx? → answer = croup (laryngotracheobronchitis) caused by parainfluenza virus (paramyxovirus); classically "seal-like, barking cough"; neck x-ray shows steeple sign **(subglottic narrowing)**; Tx = supportive; if you're forced to pick an actual Tx however, choose "nebulized racemic epinephrine."

- 8M + two weeks ago recovered from influenza infection + now has fever + inspiratory stridor + can open mouth fully; Dx? → answer = bacterial tracheitis caused by *Staph aureus.* Dx with bronchoscopy; Tx with antibiotics. Being able to open the mouth fully is important detail, since with peritonsillar abscess, one has difficulty opening the mouth fully. If a patient has recently recovered from a viral URTI, *S. aureus* secondary infections are a HY possible sequela.
- 40M + 6-month Hx of episodic sinusitis + cough with blood-tinged sputum + no improvement of Sx after several months of antibiotics, decongestants, and nasal corticosteroids + P/E shows two small ulcerations on nasal mucosa; Dx? → answer = eosinophilic granulomatosis with polyangiitis (formerly known as Churg-Strauss) → presents as asthma-like Sx and eosinophilia in a patient with positivity for p-ANCA (anti-myeloperoxidase; anti-MPO); overlap of features with granulomatosis with polyangiitis (formerly known as Wegener, which is c-ANCA [anti-proteinase 3; anti-PR3]).
- 44M + hemoptysis + hematuria + ANCA screen is negative; next best step? → answer = anti-glomerular basement membrane antibody screen → Dx likely Goodpasture; if anti-GBM antibodies positive, do renal biopsy showing linear immunofluorescence; anti-GBM Abs = antibodies against type IV collagen ("2, 3, 4; 2, 3, 4; 2, 3, 4 – The Goodpasture is marching in the field! - 2, 3, 4; 2, 3, 4; 2, 3, 4" → type II sensitivity against the alpha-3 chains of type IV collagen).
- How to Dx TB?
    - PPD skin test is performed first diagnostically. If history of BCG vaccine, do interferon-gamma release assay (IGRA) instead. **Do not do IGRA in addition to PPD.**
    - If PPD is negative, repeat after one week. If negative again, no further studies indicated. Repeats performed within 1 week may cause a false (+) secondary to a "booster reaction."
    - If IGRA is negative, no further studies indicated.
    - If PPD or IGRA is positive, do CXR. **Do not repeat positive PPD tests.**
    - If CXR is negative, treat for latent TB / give TB prophylaxis. On the USMLE, "treatment for latent TB" and "administer TB prophylaxis" mean the same thing.
    - If CXR is positive, treat for active TB.
- **How to manage a positive PPD test?**
    - Measure induration only. **Erythema does not count.**
- 5+ mm

  - Recent contact with people with active TB
  - HIV + status
  - Organ transplant recipients
  - Chronic prednisone use (>15mg/day for >1 month); anti-TNF-α agent use
  - Findings consistent with TB on CXR
- 10+ mm
  - **Immigrant status** (Western countries not included)
  - IV drug users
  - Healthcare workers; prison workers; homeless shelter personnel
  - TB laboratory personnel
  - Children under 4 years of age
- 15+ mm
  - Everyone
- How to Tx latent TB?
  - **9 months INH + pyridoxine (vitamin B6) - The USMLE Steps 1 and 2CK assess this as the answer.**
  - 4 months rifampin
  - 3 months INH + rifapentine + pyridoxine
  - Vitamin B6 must be given with INH to prevent vitamin B6 deficiency.
- **Tx of active TB**
  - **Rifampin, INH, pyrazinamide, ethambutol (RIPE) for 2 months, followed by RI alone for 4 more months (6 months total)**
  - And of course add pyridoxine (annoying that it sounds similar to pyrazinamide)
- 32M + fever of 101F + CXR shows bilateral interstitial infiltrates; Dx? → answer = *Mycoplasma pneumoniae*; most common cause of atypical pneumonia after viruses; classically bilateral.
- 32M + fever 101F + CXR shows lobar consolidation + dullness to percussion on P/E; Dx? → answer = *Streptococcus pneumoniae* → need to know is gram-positive diplococci (don't confuse with *Neisseria gonorrhea* and *meningitides*, which are gram-*negative* diplococci).

- 32M + fever 101F + CXR shows lobar consolidation with interstitial markings; Dx? → answer = *Mycoplasma* (on one of the 2CK NBMEs; Strep pneumo wasn't listed) → likely implication is that even though Strep pneumo is classically lobar and Mycoplasma bilateral, if the Q says the word "interstitial," that wins over distribution/location.
- Pneumonia + diarrhea + hyponatremia; Dx? → answer = *Legionella pneumophila.*
- Pneumonia in a cattle farmer; Dx? → answer = *Coxiella burnetti.*
- Pneumonia in someone with a pet rabbit; Dx? → answer = *Francisella tularensis.*
- Pneumonia in bird owner; Dx? → answer = *Chlamydia psittaci.*
- Pneumonia in 2-week-old neonate who had ophthalmia neonatorum treated a week ago; Dx? → answer = *Chlamydia trachomatis* pneumonia → chlamydial neonatal conjunctivitis → drains through nasolacrimal duct into nasopharynx and into the lung.
- Pneumonia in someone who recently recovered from viral infection (usually flu); Dx? → answer = *Staph aureus* → similar to post-viral bacterial tracheitis caused by *Staph.*
- Lobar pneumonia in HIV patient; Dx? → answer = *Strep pneumo*, not *Pneumocystis.*
- HIV patient + CXR shows bilateral ground-glass pneumonia; next best step in Dx? → answer = bronchoalveolar lavage with silver-staining of the yeast; Dx? → *Pneumocystis jirovecii* → Tx = TMP/SMX.
- Empiric Tx for community-acquired pneumonia (CAP)? → answer = azithromycin (macrolide) → covers atypicals (i.e., *Mycoplasma,* etc.) and *S. pneumo.*
- Empiric Tx for CAP when patient's had antibiotics in the past three months? → answer = respiratory fluoroquinolone (i.e., levofloxacin).
- How do we Tx CAP if admitting patient to hospital?
    - If non-ICU: fluoroquinolone, OR beta-lactam + macrolide.
    - If ICU: beta-lactam + EITHER macrolide or fluoroquinolone.
    - **If septic, ceftriaxone +/- vancomycin frequently given** → ceftriaxone common singular Abx choice in sepsis; very effective against *S. pneumo* in particular; ceftriaxone as monotherapy is all over the 2CK-level NBMEs for Tx of sepsis (not just CAP); ceftriaxone + vancomycin is a combo is the answer on one of the newer forms for septic patient with CAP (increasing

resistance of *S. pneumo* to ceftriaxone); if patient is child under 6, give cefotaxime instead of ceftriaxone (less displacement of bilirubin from albumin; HY for 2CK NBMEs).

- When is pneumonia considered hospital-acquired (HAP)? → if Sx start >48 hours after admission to hospital.
- How is HAP and ventilator-acquired pneumonia (VAP) Tx differently from CAP? → **need to cover MRSA and *Pseudomonas* just in case** → Tx entails various broad-spectrum options; classics are **vancomycin + third- or fourth-generation cephalosporin;** ceftazidime (3rd gen ceph) and cefepime (4th gen ceph) are effective against *Pseudomonas;* **piperacillin/tazobactam** ("PipTaz") is classic combo; amikacin (aminoglycoside) is effective against *Pseudomonas*; ceftaroline and ceftobiprole are 5th gen cephs effective against MRSA; meropenem (or imipenem/cilastatin) also used for HAP/VAP.
- First-line Tx in COPD? → SABA or SAMA (short-acting muscarinic antagonist; ipratropium) → answer on one of the 2CK NBMEs is just straight-up ipratropium (home oxygen and pulmonary rehabilitation are wrong answers for first-line Tx).
- When to do home oxygen therapy in COPD? → when patient's arterial pO2 <55 mmHg (or <60 mmHg if patient has cor pulmonale).
- 55M with COPD; number-one way to decrease mortality? → answer = smoking cessation; if patient has already stopped smoking, answer = home oxygen therapy (decreases mortality, but only indicated as per criteria above).
- 48M + BMI 45 + snores loudly during sleep; what is most likely seen in this patient? → answer = increased serum bicarbonate → Dx = obesity hypoventilation syndrome → blood levels of CO2 are increased, with bicarb increased to compensate (chronic respiratory acidosis).
- 8-month-old girl + stridor that improves with neck extension; Dx? → answer = vascular ring → weird but HY diagnosis for peds → aberrant embryologic development where the aorta and/or surrounding vessels form a ring around the esophagus and/or trachea.
- 8-month-old girl + stridor that improves when prone or upright; Dx? → laryngomalacia → most common cause of stridor in peds → soft cartilage of upper larynx collapses during inhalation.
- 3F + 2-wk Hx of cough and nasal congestion + snoring loudly past 6 months + P/E shows she breathes predominantly through her mouth + 1/6 holosystolic murmur and loud S2 + CXR shows cardiomegaly + increased pulmonary vascular markings + echo shows RV hypertrophy and mild tricuspid regurg;

what is the most appropriate long-term Mx for this patient? → answer = adenoidectomy and tonsillectomy → sounds weird, but HY for Peds shelf → can cause obstructive lung disease with cor pulmonale → loud S2 and increased pulmonary vascular markings suggest pulmonary HTN; tricuspid regurg can be seen sometimes in pulmonary HTN.

- 2F + stridor + laryngoscopy shows small growths of larynx; Dx? → answer = HPV 6/11 → laryngeal papillomatosis.
- Highest yield points about cystic fibrosis? → autosomal recessive; chromosome 7; *CFTR* gene; codes for chloride channel that functions to secrete chloride in the lungs and pancreas, and reabsorb chloride in the sweat glands; ΔF508 (deletion of phenylalanine at position 508) is most common mutation; sweat chloride test >60 mEq/L is most diagnostic (more than genotyping); mutated CFTR channel is usually retained in rough endoplasmic reticulum in the cytosol (i.e., doesn't make it to cell surface); disease causes a negative transepithelial potential difference (TEPD) across nasal epithelium; neonates can be screened with a positive blood immunoreactive trypsinogen; *Pseudomonas* eclipses *S. aureus* as most common cause of pneumonia after age 10; before age 10, *S. aureus* eclipses *Pseudomonas*; male infertility due to congenital bilateral absence of vas deferens (CBAVD); various Txs exist, however a couple to be aware of: dornase-alfa is a deoxyribonuclease that can help reduce the viscosity of mucous secretions; Ivacaftor is a CFTR potentiator that helps restore function of the misfolded protein.
- 32M + Hx of recurrent lung infections + two years of inability to have children with wife + wife has two children from prior marriage + his sperm sample shows immotile sperm; Dx? → answer = Kartagener syndrome (primary ciliary dyskinesia) → dynein arm defect of cilia (a cilium on cross-section has a 9x2 arrangement of microtubules); associated with dextrocardia / situs inversus; sperm are immotile (require cilia function); **this contrasts with CF, which has no sperm in the sample due to CBAVD.**
- What should I know about breath sounds, percussion, tracheal shift, etc.? Basically all of that annoying physical exam stuff."

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Routine Respiratory Exam Findings				
Scenario	Percussion	Tactile fremitus	Tracheal shift	Breath sounds
Healthy lung	Resonance	Normal	None	Vesicular
Pleural effusion	Stony dull	Decreased	Away (if large)	Decreased
Pneumothorax	Hyper-resonant	Decreased	Away (tension) Toward (spontaneous)	Decreased
Atelectasis	Dull	Decreased	Toward (if large)	Decreased
Pneumonia	Dull	Increased	None	Bronchial
Emphysema	Hyper-resonant	Decreased	None	Decreased

- Pleural effusion in someone who has pneumonia; Dx? → answer = parapneumonic effusion (exudative pleural effusion secondary to a lung infection).
- What is empyema? → pus in a preexisting cavity (i.e., refers to pus in the pleural space); in contrast, an abscess is pus in a location where there was not a preexisting cavity (e.g., the forearm).
- Characteristic of parapneumonic effusion most predictive of progression to empyema? → answer = decreased pH of the pleural fluid (<7.1).
- Transudative vs exudative pleural effusion? → Light's criteria for exudative:
    - **Pleural fluid to serum protein ratio >0.5,** OR
    - **Pleural fluid to serum LDH ratio >0.6,** OR
    - Pleural fluid LDH > 2/3 upper limit of serum LDH.
- Any other characteristics notable for transudative vs exudative? → qualitatively, in exudative, pleural fluid cell count will be higher than in transudative; glucose can also be reduced in infective causes (parapneumonic).
- 44M + long Hx of smoking + dullness to percussion 2/3 up the lung field; Dx? → answer = malignant pleural effusion → malignancy, congestive heart failure, and infection are very common causes of pleural effusion.

- 32F + ovarian fibroma + ascites + hydrothorax; Dx? → answer = Meig syndrome.
- Treatment for pulmonary embolism? → answer = heparin *before* spiral CT of the chest; if pregnant, do V/Q scan instead of CT (sufficient for USMLE).
- Treatment for pulmonary embolism? → answer = heparin *before* spiral CT of the chest; if pregnant, do V/Q scan instead of CT (sufficient for USMLE).
- When is IVC filter the answer for pulmonary embolism? → if the patient gets a PE while *already* on anticoagulation (warfarin, dabigatran, etc.) *and* the spiral CT has already been performed. In other words:
    - 58F on warfarin + gets PE; next best step? → answer = CT.
    - 58F on warfarin + gets PE + CT confirms Dx; next best step? → answer = IVC filter.
- 32M + plays basketball + few hours of shortness of breath; Dx + Tx? → spontaneous pneumothorax caused by ruptured subapical bleb; Tx = needle decompression followed by chest tube (some small pneumothoraces can be observed, but on USMLE always Tx).
- 32M + MVA + dyspnea + low BP + breath sounds decreased on left + tracheal shift to right; Dx + Tx? → answer = tension pneumothorax; Tx = needle decompression followed by chest tube.
- "Cephalization of pulmonary vasculature"; Dx? → pulmonary edema → important description that shows up in NBME vignettes → due to increased hydrostatic pressure from left heart pathology.
- 36M + long-bone fractures + petechiae on the chest; Dx? → answer = fat embolism.
- Motor vehicle accident (MVA) + paradoxical breathing (chest moves outward with exhalation; inward with inhalation); Dx? → answer = flail chest.
- MVA + rib fractures + underlying infiltrates in lung + low O2 sats; Dx? → answer = pulmonary contusion.
- MVA + no rib fractures + non-central chest pain + pulmonary infiltrates underlying the painful area; Dx? → answer = pulmonary contusion (resources will say "white out of the lung" for pulmonary contusion, but this is buzzywordy and never shows up on actual NBME material).
- MVA + pulmonary infiltrates + low O2 sats + bolus of normal saline given, resulting in worsening of O2 sats; Dx? → answer = pulmonary contusion (contused lung is very sensitive to fluid overload).
- MVA + **bruising/pain over the sternum** +/- rib fractures; Dx? → answer = myocardial contusion.

- MVA + bruising/pain over sternum + pulmonary infiltrates + O2 sats get worse when saline is given; Dx? → answer = myocardial contusion ("Wait, but I thought you said that latter finding means pulmonary contusion") → it does, and it's HY for pulmonary contusion, but "bruising/pain over the sternum" wins if it's listed; this is on a 2CK NBME.
- Important point about Mx of myocardial contusion? → do troponins + must monitor for arrhythmia.
- Adult male + abdo pain + Hx of alcohol use + diffuse pulmonary infiltrates + low O2 sats; Dx? → answer = ARDS → must have pO2/FiO2 <300; if the Q asks about ventilator settings, know that **low-tidal volume mechanical ventilation** is often used (prevents barotrauma) + permissive hypercapnia + prone positioning.
- When you get a random ventilator Q and they want an answer? → "increase PEEP" almost always right.
- Patient has improving O2 sats on ventilator; next best step? → "wean from ventilator."
- 18M + Hx of asthma + 2-day Hx of right cheek tenderness + Hx of several pneumonias and sinusitis occurrences; Dx? → answer = IgA deficiency → answer will often be listed as "impaired humoral immunity" or "deficiency of mucosal immunoglobulin"; sore cheek = classic for sinusitis; presents as recurrent sinopulmonary infections; also associated with **Hx of Giardia infection, autoimmune diseases (e.g., vitiligo), and atopy** (dry cough in winter [cough-variant asthma], hay fever in spring, eczema in summer); anaphylaxis with blood transfusion is "too easy" for most 2CK IgA deficiency Qs but will rarely show up, yes.
- How to differentiate viral from bacterial upper respiratory tract infection (URTI)? → CENTOR criteria:
    - If 0 or 1 point, the URTI is unlikely to be bacterial (i.e., it's likely to be viral). If 2-4 points, chance is much greater that URTI is bacterial.
    - **1) <u>Absence of</u>** cough (i.e., no cough = 1 point; if patient has cough = 0 points).
    - **2)** Fever.
    - **3)** Tonsillar exudates.
    - **4)** Lymphadenopathy (cervical, submandibular, etc.).
- There is a version of the criteria that includes age, but on the USMLE it can cause you to get questions wrong. So just use the simplified above four points.

- If 0-1 point, answer = “supportive care”; or “no treatment necessary”; or “warm saline gargle” (same as supportive care); or “acetaminophen.” Latter is answer for 3M with viral URTI + fever on Peds NBME form 2.
- If 0-2 points, next best step = “rapid Strep test.” If rapid Strep test is negative, answer = throat culture, **NOT** sputum culture.
- While waiting on the throat culture results, we send the patient home with amoxicillin or penicillin for presumptive Strep pharyngitis.
- If child is, e.g., 12 years old, and develops a rash with the beta-lactam, answer = beta-lactam allergy.
- If the vignette is of a 16-17 year-old who has been going on dates recently (there will be no confusion; the USMLE will make it clear), the answer = EBV mononucleosis; therefore do a heterophile antibody test (Monospot test).
- **EBV is the odd virus out that usually presents with all four (+) CENTOR criteria.**
- This is why it’s frequently misdiagnosed as Strep pharyngitis. It is HY to know that beta-lactams given to patients with EBV may cause rash via a hypersensitivity response to the Abx in the setting of antibody production to the virus. EBV, in a patient who does not receive Abx, can cause a mild maculopapular rash. But the rash with beta-lactam + EBV causes a more intense pruritic response generally 7-10 days following Abx administration on the extensor surfaces + pressure points.

- Which parameters shift the Hb-O2 dissociation curve to the right? → ↑ temperature; ↑ 2,3-BPG; ↑ CO2; ↓ pH; ↑ H^+ → right-shift means increased oxygen unloading at tissues; protons in the blood are buffered by deoxygenated hemoglobin (Step 1 NBME).
- 50M + arterial pO2 normal + arterial O2 content low; why the latter? → answer = anemia; arterial O2 content = amount of O2 dissolved in blood (pO2) + amount of O2 bound to Hb (Hb saturation); if pO2 is normal, then it should be able to bind to Hb just fine; therefore likely way O2 content is still low is if Hb levels are low (anemia).



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